QUESTION 1

What roles are essential from the business side in a data warehouse project?

Answer

Essential roles include business sponsor, business driver, business lead, and business users, all crucial for project support and user acceptance.

QUESTION 2

What factors should be considered in the product selection process?

Answer

Factors include functionality, technical architecture compatibility, vendor viability, and the impact on existing infrastructure.

QUESTION 3

What is meant by 'fact table granularity'?

Answer

Fact table granularity refers to the level of detail stored in a fact table, determining how much data is aggregated and what dimensions are included.

QUESTION 4

What role does the data warehouse play in CRM?

Answer

The data warehouse serves as the central repository for integrating and storing customer data from various operational systems, enabling a comprehensive view of customer interactions.

QUESTION 5

What are typical attributes found in a 'customer dimension'?

Answer

Typical attributes include Customer ID, Name, Address, Email, and Phone, which help in segmenting and analyzing customer data for targeted marketing.

QUESTION 6

What is a Product Key (PK)?

Answer

A Product Key (PK) is a unique identifier for each product in a dimensional model, ensuring that each product can be distinctly referenced.

QUESTION 7

How can SQL queries leverage the date dimension table for analysis?

Answer

SQL queries can join the date dimension to filter and slice data by attributes like weekdays, holidays, and fiscal periods, enhancing reporting capabilities.

QUESTION 8

What does the term 'set difference' refer to in analyzing sales?

Answer

Set difference is used to identify products that were on promotion but did not sell, comparing lists from the promotion coverage and POS sales fact tables.

QUESTION 9

How does a data warehouse project align with business strategy?

Answer

A data warehouse project should align with business strategy by addressing key business needs and providing insights that support informed decision-making and strategic goals.

QUESTION 10

What is Customer Relationship Management (CRM)?

Answer

CRM is a business strategy focused on understanding and managing customer relationships to enhance customer satisfaction, retention, and profitability.

QUESTION 11

What is the impact of improved access to information in a data warehouse?

Answer

Improved access to information enhances decision-making, promotes efficiency, and can lead to quantifiable business benefits.

QUESTION 12

What types of transactions are generated in claims processing?

Answer

Transactions include opening, reopening, and closing claims; setting, resetting, and closing reserves; adjusting salvage estimates; conducting adjuster inspections; opening and closing lawsuits; making and receiving payments; and subrogating claims.

Flashcards 13 - 24

QUESTION 13

What does the term 'data warehousing' encompass?

Answer

Data warehousing refers to the process of publishing the right data for organizational needs, integrating various data sources for analysis and decision-making.

QUESTION 14

What types of promotions can a promotion dimension track?

Answer

The promotion dimension can track temporary price reductions, end-aisle displays, newspaper ads, and coupons, reflecting the various conditions affecting product sales.

QUESTION 15

What is the importance of a clear allocation structure?

Answer

A clear allocation structure helps streamline the data warehousing process and ensures that financial data is accurate and reliable for analysis.

QUESTION 16

Why is it important to have a comprehensive set of facts in a surgical events transaction fact table?

Answer

A comprehensive set of facts enables thorough P&L analysis for surgical procedures by allocating various costs to each surgical event.

QUESTION 17

Define the term 'premium snapshot.'

Answer

A premium snapshot is a summary of premium revenue associated with each policy over time, essential for analyzing profitability.

QUESTION 18

What are the typical methods for session tracking in web applications?

Answer

Session tracking methods include using cookies, time-contiguous log entries, SSL logins, hidden fields in URLs, and persistent cookies.

QUESTION 19

What occurs during the investigative phase of a claim?

Answer

The insurance company sends an adjuster to examine the covered item and interview the claimant or other involved parties, producing a stream of transactions related to the claim.

QUESTION 20

What is a Universal Product Code (UPC)?

Answer

A Universal Product Code (UPC) is a standard bar-coded value found on most grocery and drug store merchandise used for inventory tracking.

QUESTION 21

How can performance tuning be achieved in a data warehouse?

Answer

Performance tuning can be achieved through strategies such as indexing, creating aggregate tables, and optimizing query designs.

QUESTION 22

Why is it important to establish standards for analytic applications?

Answer

Establishing standards ensures consistency in user experience, simplifies development, and enhances communication between developers and business users.

QUESTION 23

What is the impact of using accumulating snapshot facts on billing analysis?

Answer

Using accumulating snapshot facts allows for a comprehensive view of the billing timeline, enabling detailed analysis of payment processes and outstanding balances.

QUESTION 24

What are present imperatives in data warehousing?

Answer

Present imperatives include addressing ongoing technology advances, managing data privacy, and ensuring effective data governance to prevent misuse of data.

Flashcards 25 - 36

QUESTION 25

Define the term 'policyholder' in the context of insurance.

Answer

A policyholder is the individual or entity that owns an insurance policy and is responsible for paying the premiums.

QUESTION 26

What is a minidimension in CRM?

Answer

A minidimension is a separate dimension created to hold frequently changing or analyzed attributes, simplifying performance and maintenance in large customer dimensions.

QUESTION 27

What are the challenges with using UNION and INTERSECTION in SQL queries?

Answer

Using UNION can simplify OR queries, but INTERSECTION can complicate AND queries due to SQL's difficulties in handling constraints across multiple rows, often requiring custom solutions.

QUESTION 28

How does multi-currency support operate in order management?

Answer

Multi-currency support involves capturing transactions in local currencies and converting them to a standardized corporate currency, facilitating accurate reporting across different regions.

QUESTION 29

What is the significance of having a usage monitoring system in a data warehouse?

Answer

A usage monitoring system aids in ongoing performance tuning, user support, capacity planning, and helps justify resource allocation.

QUESTION 30

What is the significance of the product dimension in retail sales?

Answer

The product dimension contains essential details about products, including descriptions, categories, and attributes necessary for sales reporting.

QUESTION 31

What is a factless fact table?

Answer

A factless fact table records events or transactions without associated numeric measures, often used to track occurrences of non-numeric events.

QUESTION 32

What is a Value Chain?

Answer

A Value Chain is a sequence of processes that describe how products or services move from creation to final sales.

QUESTION 33

What are periodic snapshots in the context of the general ledger?

Answer

Periodic snapshots capture the state of the general ledger accounts at the end of each fiscal period, providing a historical view of financial positions over time.

QUESTION 34

What are some key attributes in the policyholder dimension?

Answer

Key attributes include name, contact information, demographic details, and policy history.

QUESTION 35

What does churn mean in subscription services?

Answer

Churn refers to the rate at which customers discontinue their subscriptions, measured by the number of customers lost versus gained over a specific period.

QUESTION 36

What are lag calculations in the context of order management?

Answer

Lag calculations measure the time spans between major milestones in the order fulfillment process, helping analyze the efficiency of operations.

Flashcards 37 - 48

QUESTION 37

What is the purpose of the skills group keyword dimension?

Answer

It allows for the categorization of technical skills associated with employees, facilitating querying based on those skills.

QUESTION 38

How does inventory management contribute to operational efficiency?

Answer

Effective inventory management ensures that the right amount of stock is available at the right time, minimizing costs while meeting customer demand.

QUESTION 39

What is dimensional modeling?

Answer

Dimensional modeling is a design technique used in data warehousing that organizes data into fact and dimension tables to facilitate easier data retrieval and reporting.

QUESTION 40

What steps should be taken immediately after collecting business requirements?

Answer

Debrief the interview team, document insights, and create a findings document that organizes requirements by key business processes.

QUESTION 41

What does GUI stand for?

Answer

Graphic User Interface, characterized by windows, icons, graphics, and mouse pointing devices.

QUESTION 42

What are the key attributes of a bridge table?

Answer

A bridge table typically includes the parent customer key, subsidiary customer key, the number of levels from the parent, and flags to indicate whether a customer is a top-most or bottom-most entity.

QUESTION 43

What is a transaction in a database?

Answer

A transaction is an indivisible unit of work in a transaction processing system that must be completed in its entirety or not at all.

QUESTION 44

What is the purpose of a 'decision support system' (DSS)?

Answer

A decision support system (DSS) is designed to assist in making business decisions by analyzing and presenting data to support decision-making processes.

QUESTION 45

What is the role of a data steward in a data warehouse project?

Answer

A data steward ensures enterprise agreement on conformed dimensions and facts, managing the quality and consistency of data across the warehouse.

QUESTION 46

Why is it important to declare the grain in dimensional modeling?

Answer

Declaring the grain ensures all team members agree on the level of detail, preventing inconsistencies and issues in data warehouse implementation.

QUESTION 47

What is the role of a bridge table in hierarchical customer structures?

Answer

A bridge table facilitates the analysis of revenue results at any hierarchical level by connecting customer dimensions to fact tables, allowing for more complex queries.

QUESTION 48

What is the grain of the first fact table in a clickstream data mart?

Answer

The grain of the first fact table is one row for each completed customer session, which allows for manageable data analysis while capturing essential session information.

Flashcards 49 - 60

QUESTION 49

What is the significance of the data warehouse bus architecture?

Answer

The bus architecture integrates various business processes into a cohesive data warehouse, using common dimensions and facts across dimensional models.

QUESTION 50

What strategies can mitigate catastrophic failures in data warehouses?

Answer

Mitigation strategies include implementing distributed systems, maintaining redundant communication paths, using secure storage solutions, and ensuring robust access controls.

QUESTION 51

Why is allocation critical in data warehousing?

Answer

It enables the ability to slice and dice order facts by dimensions, particularly product, which is essential for comprehensive analysis.

QUESTION 52

What are some attributes commonly included in a date dimension table?

Answer

Common attributes include date key, full date description, day of week, calendar month, calendar year, fiscal year-month, holiday indicator, and weekday indicator.

QUESTION 53

What is the significance of the 'Utilization Status' dimension in facilities management?

Answer

The 'Utilization Status' dimension indicates whether a facility is available or in use, aiding in analysis of facility management and optimization.

QUESTION 54

Explain the concept of 'data quality assurance'.

Answer

Data quality assurance is the step in the data staging process where data is tested for consistency, completeness, and fitness before being published to users.

QUESTION 55

What is the purpose of the glossary provided in the book?

Answer

The glossary serves as a reference to bridge the gap between general business understanding and the specific terminology used in the case studies.

QUESTION 56

What is a POS transaction?

Answer

A POS transaction refers to a sale made at the point of sale, where customer purchases are recorded and analyzed.

QUESTION 57

Why is it important to track dimension attribute changes?

Answer

Tracking changes in dimension attributes helps users understand the impact of updates on data over time, offering insights into trends and variations in the data.

QUESTION 58

What is the role of the Profiler in clickstream data?

Answer

The Profiler provides demographic information about web visitors and can help in personalizing the user experience by using cookies to identify returning visitors.

QUESTION 59

What is the role of a special join key in data modeling?

Answer

A special join key uniquely identifies a single account in a specific month, facilitating accurate joins between core fact tables and custom extended fact tables.

QUESTION 60

Define Point-of-Sale (POS) System.

Answer

The cash registers and associated in-store computers used in a retail environment to process sales transactions.

Flashcards 61 - 72

QUESTION 61

What is the role of market research in product selection?

Answer

Market research helps to understand the offerings of various vendors and aids in making informed purchasing decisions.

QUESTION 62

What quality assurance techniques should be applied to attribute values in dimensional models?

Answer

Quality assurance should include checking for misspellings, impossible values, and ensuring that attribute values are completely populated. Sorting distinct values can help identify minor spelling differences.

QUESTION 63

What is the purpose of a medical record entries fact table?

Answer

The medical record entries fact table captures various types of medical measurements and records for patients, allowing for analysis and reporting.

QUESTION 64

Explain what atomic data means in a data mart.

Answer

Atomic data refers to the most granular level of detail in a data mart, which is necessary to support detailed user queries and ad hoc analysis.

QUESTION 65

How can organizations improve access to financial data?

Answer

By implementing robust data warehouse solutions that centralize financial information, organizations can enhance data accessibility for analysis and reporting.

QUESTION 66

What is the purpose of data staging in a data warehouse?

Answer

Data staging prepares raw data from operational systems for the dimensional model in the data presentation area.

QUESTION 67

What are Minidimensions?

Answer

Subsets of a large dimension broken into smaller dimensions to control rapid growth, often for demographic attributes.

QUESTION 68

What is the purpose of the audit dimension in dimensional modeling?

Answer

The audit dimension tracks data lineage and quality metadata within a fact table, allowing for transparency and accountability in data processing.

QUESTION 69

What is a Schema in database terms?

Answer

The logical or physical design of a set of database tables, indicating the relationships among the tables.

QUESTION 70

What are dimension outriggers?

Answer

Dimension outriggers are tables that contain low-cardinality attributes linked to a primary dimension, allowing for better organization and space savings without fully snowflaking the dimension.

QUESTION 71

Explain the concept of Index in databases.

Answer

A data structure associated with a table that improves database performance and query access speed, typically using B-tree or bitmap indexes.

QUESTION 72

What does the grain of the invoice fact table represent?

Answer

The grain of the invoice fact table is defined as one row per individual invoice line item, allowing for detailed analysis of each shipment.

Flashcards 73 - 84

QUESTION 73

What is the importance of a single integrated view of each customer in CRM?

Answer

A single integrated view enables organizations to maintain long-lasting relationships with customers by ensuring consistency in interactions and data across all departments.

QUESTION 74

What is an account-to-customer bridge table?

Answer

It is a table that links accounts to multiple customers while maintaining the granularity of the dimension table.

QUESTION 75

Why is activity-based costing important in data warehousing?

Answer

Activity-based costing provides a more accurate allocation of costs to products and services, enhancing the reliability of financial data for decision-making.

QUESTION 76

What are some typical sources of customer data in a data warehouse?

Answer

Typical sources include transactional data, interaction data, demographic and behavioral data, and self-provided profile data.

QUESTION 77

What are the major activities in building a data warehouse?

Answer

Major activities include project planning, requirement gathering, dimensional modeling, data staging, and ensuring data integrity and security throughout the process.

QUESTION 78

What is the impact of too many dimensions in a fact table?

Answer

Having too many dimensions can increase disk space requirements and complicate queries, making it harder to analyze data effectively.

QUESTION 79

Define 'Mortgage-Specific Account Key (PK)'.

Answer

A Mortgage-Specific Account Key (PK) is a unique identifier used specifically for mortgage accounts, distinguishing them from other account types.

QUESTION 80

Describe a Transaction Fact Table.

Answer

A Transaction Fact Table captures individual transaction events at a specific point in time, with a row existing only if a transaction event occurred. It allows detailed analysis of customer or product behavior.

QUESTION 81

What is a market basket analysis?

Answer

Market basket analysis analyzes the combinations of products sold together, providing insights for merchandising and marketing strategies.

QUESTION 82

How do data marts relate to business processes?

Answer

Data marts are derived from specific business processes and correspond to operational source systems, allowing focused data analysis based on business activities.

QUESTION 83

Define the budgeting process in financial management.

Answer

The budgeting process involves creating, approving, and adjusting budgets throughout the fiscal year to reflect changing business conditions and actual spending.

QUESTION 84

What is the importance of the Event Dimension?

Answer

The Event Dimension describes what happened on the page during a customer interaction, providing crucial insights into user behavior and engagement.

Flashcards 85 - 96

QUESTION 85

What should be prepared before gathering business requirements?

Answer

Prepare a forum for meetings, identify the requirements team, and create a questionnaire to organize thoughts and guide the discussion.

QUESTION 86

What is the AND/OR dilemma in querying technical skill-set keywords?

Answer

The dilemma arises when trying to satisfy queries that require both conditions (AND) versus either condition (OR). SQL struggles with AND constraints across multiple rows, complicating data retrieval.

QUESTION 87

What does the term 'policy transactions' refer to?

Answer

Policy transactions encompass all transactions related to insurance policies, including premiums, claims, and policyholder details.

QUESTION 88

What challenges do telecommunications and utilities face in data modeling?

Answer

Challenges include managing geographic data, ensuring data integrity across multiple dimensions, and creating models that accommodate diverse service offerings.

QUESTION 89

What are off-invoice allowances?

Answer

Off-invoice allowances are discounts or allowances applied to line items that are not shown on the customer invoice but are relevant for internal accounting.

QUESTION 90

What is a periodic snapshot fact table in inventory management?

Answer

A periodic snapshot fact table records inventory levels at regular intervals, providing a historical perspective of inventory balances, often used to analyze inventory movements over time.

QUESTION 91

What is a data mart?

Answer

A data mart is a logical and physical subset of a data warehouse, originally defined as highly aggregated data to answer specific business questions.

QUESTION 92

Why would dimensions be combined into a superdimension?

Answer

When data volumes are small or when a need arises for additional attributes that depend on the combined context and meaning of the dimensions.

QUESTION 93

What is a Periodic Snapshot Fact Table?

Answer

A Periodic Snapshot Fact Table captures cumulative performance metrics at regular, predictable time intervals, allowing trend analysis over time.

QUESTION 94

What are the implications of using a 4-byte integer for surrogate keys?

Answer

A 4-byte integer can accommodate approximately 2 billion positive values, optimizing data storage and performance in the fact table.

QUESTION 95

Explain the concept of GMROI.

Answer

GMROI, or Gross Margin Return on Investment, measures the effectiveness of inventory investment by evaluating the profit generated per dollar invested in inventory.

QUESTION 96

How does drilling down work in a data mart?

Answer

Drilling down involves adding row headers from dimension tables to provide more detailed information, enabling users to analyze data at finer levels of granularity.

Flashcards 97 - 108

QUESTION 97

How are sales dollar amounts calculated in the retail schema?

Answer

Sales dollar amounts are calculated by summing the sales dollar amounts from the retail sales fact table grouped by relevant dimensions.

QUESTION 98

What is the importance of maintaining historical data in dimensional structures?

Answer

Maintaining historical data allows users to analyze trends over time, making it essential for comprehensive business analysis.

QUESTION 99

What is a consolidated fact table?

Answer

A consolidated fact table combines metrics from different business processes, allowing for easier comparison and analysis of actual versus budgeted amounts.

QUESTION 100

What are the implications of customer segmentation attributes in CRM?

Answer

Customer segmentation attributes allow businesses to classify customers based on behaviors and characteristics, aiding in targeted marketing and service strategies.

QUESTION 101

What is the importance of data cleansing in customer data consolidation?

Answer

Data cleansing is crucial for removing duplicates and correcting inaccuracies in customer data, which helps to maintain data integrity and improves the effectiveness of CRM initiatives.

QUESTION 102

Explain 'Turns' in inventory management.

Answer

Turns refer to the number of times inventory is completely replenished in a given period, usually a year, to keep pace with sales rates.

QUESTION 103

What is the purpose of a shipper dimension in an invoice fact table?

Answer

The shipper dimension describes the delivery method and carrier used to transport products, allowing analysis of shipping performance and costs.

QUESTION 104

What are the dimensions associated with the payments fact table?

Answer

The dimensions associated with the payments fact table include the commitment dimensions plus a payment dimension to identify the type of payment and the payee.

QUESTION 105

What is a surrogate key in dimensional modeling?

Answer

A surrogate key is a unique identifier used in a dimensional model that is not derived from application data. It helps avoid issues arising from duplicate operational product keys and tracks changing product attributes.

QUESTION 106

What is a claims accumulating snapshot?

Answer

A claims accumulating snapshot is a periodic update of all transactions related to a claim, allowing for performance measures to be derived without traversing every detailed transaction.

QUESTION 107

What are the present imperatives for data warehousing?

Answer

Present imperatives include the need for real-time data access, improved data quality, integration of diverse data sources, and enhanced analytical capabilities.

QUESTION 108

What is a bridge table in dimensional modeling?

Answer

A bridge table is an associative table used to manage relationships in variable-depth hierarchies, allowing proper SQL queries to summarize data across hierarchical levels without modifying the customer dimension or fact tables.

Flashcards 109 - 120

QUESTION 109

What are conformed dimensions?

Answer

Conformed dimensions are dimensions that are consistent across different data marts, enabling users to compare data from various business processes seamlessly.

QUESTION 110

What is scope creep in data warehouse projects?

Answer

Scope creep refers to the gradual expansion of project scope due to the accumulation of minor changes, which can significantly alter the project's original goals and timelines.

QUESTION 111

What is the purpose of a fact dimension in a health care data warehouse?

Answer

A fact dimension organizes sparse measurements and allows for effective analysis of heterogeneous data in health care.

QUESTION 112

What are the key factors in determining the physical design of a data warehouse?

Answer

Key factors include performance tuning, indexing strategies, partitioning, and data types.

QUESTION 113

What is the significance of understanding user needs in data warehousing?

Answer

Understanding user needs is crucial for designing effective data warehousing solutions that deliver relevant insights and improve decision-making processes.

QUESTION 114

How can organizations move beyond rear-view metrics in order management?

Answer

Organizations can incorporate leading indicators that predict future order activity, supplementing historical data with proactive analytics.

QUESTION 115

What is market basket analysis?

Answer

Market basket analysis is a data mining technique used to identify patterns of items that frequently co-occur in transactions, helping businesses understand customer purchasing behavior.

QUESTION 116

What is the significance of the accounting period dimension?

Answer

The accounting period dimension allows for tracking financial data across different time frames, including actual and budget amounts.

QUESTION 117

Why is it recommended to store conversion factors in the fact table?

Answer

It is recommended to store conversion factors in the fact table to minimize user error and ensure that the correct factor is used, reducing the burden on business users and making the design more robust.

QUESTION 118

What are the risks associated with data warehouse outsourcing?

Answer

Risks include potential service disruptions if the provider goes out of business, lack of customization, and security concerns regarding data management and access.

QUESTION 119

What is the Treatment Date Key in a healthcare data model?

Answer

The Treatment Date Key is a foreign key that references the date when a particular treatment was administered to a patient.

QUESTION 120

What are the main tasks involved in fact table staging?

Answer

Main tasks include extracting fact data, ensuring dimension data is updated, separating data by granularity, and transforming facts as needed.

Flashcards 121 - 132

QUESTION 121

What is the difference between interviews and facilitated sessions for gathering requirements?

Answer

Interviews allow for individual input and easier scheduling, while facilitated sessions aim for group consensus but require more time from participants.

QUESTION 122

What is the first step in staging fact table data?

Answer

Extract fact data from the operational source system.

QUESTION 123

What are the four special dimensions found in the Web visitor clickstream data mart?

Answer

The four special dimensions are unique to the clickstream data mart and help analyze the interactions between web activities and customer behavior.

QUESTION 124

What are year-to-date facts, and why were they removed from the design?

Answer

Year-to-date facts represent cumulative totals for a year. They were removed because they can be confusing and prone to error, instead allowing users to calculate them on-the-fly using dimensions.

QUESTION 125

How can organizations manage expectations during the architecture design process?

Answer

Organizations can manage expectations by communicating the architecture plan clearly and involving stakeholders in the review and feedback process.

QUESTION 126

What does the term 'non-standard calendar calculations' refer to?

Answer

Non-standard calendar calculations include attributes like fiscal periods and holidays that are not directly supported by standard SQL date functions.

QUESTION 127

Define the term 'e-learning' in the context of data warehousing.

Answer

E-learning refers to the use of electronic technologies to access educational curriculum outside of a traditional classroom, often facilitated by data warehousing systems.

QUESTION 128

What are the implications of using weighting factors in multivalued dimensions?

Answer

Weighting factors allow for the allocation of numeric facts across multiple values but can complicate data modeling and reporting if not agreed upon.

QUESTION 129

What is the purpose of partitioning in data warehousing?

Answer

Partitioning improves query performance and speeds up data loading processes by segmenting large tables into manageable pieces.

QUESTION 130

What is a bridge table in dimensional modeling?

Answer

A bridge table is a table with a multipart key that captures a many-to-many relationship, allowing for many-valued dimensions or ragged hierarchies. It serves to connect a fact table with a dimension table.

QUESTION 131

Explain the importance of maintaining control over the data warehouse environment.

Answer

Maintaining control prevents operational changes from affecting data integrity and allows for consistent data management practices.

QUESTION 132

What defines a well-designed dimensional model?

Answer

A well-designed dimensional model is simple, intuitive for business users, and efficient for query performance, allowing easy access to meaningful data.

Flashcards 133 - 144

QUESTION 133

What are the keys to effective data warehouse project planning?

Answer

1. A solid business sponsor. 2. Balancing high value with feasibility to define the scope. 3. Collaborating with a capable team for detailed project planning. 4. Being an effective project manager by motivating and communicating across the organization.

QUESTION 134

What is the role of analytic applications in the data warehouse?

Answer

Analytic applications provide a structured framework for users to interact with data, enabling better decision-making without requiring users to start from scratch with ad hoc queries.

QUESTION 135

How does a data warehouse differ from operational systems?

Answer

A data warehouse is designed for analytical processing, allowing users to query large volumes of historical data, while operational systems focus on real-time transaction processing and record-keeping.

QUESTION 136

What factors should be considered when deciding between a single blended fact table and multiple fact tables?

Answer

Considerations include users' analytic requirements, the uniqueness of business processes, the involvement of multiple source systems, and the dimensionality of the facts.

QUESTION 137

What challenges are associated with inventory snapshot tables?

Answer

Inventory snapshot tables tend to be dense, resulting in large data sizes that may require compromises in frequency and granularity for storage efficiency.

QUESTION 138

Explain the process of market basket analysis.

Answer

Market basket analysis involves identifying which products are frequently purchased together to inform marketing strategies, product placement, and promotions.

QUESTION 139

What are the key design goals of dimensional modeling?

Answer

The key design goals of dimensional modeling are user understandability, query performance, and resilience to change.

QUESTION 140

What is a clickstream?

Answer

A clickstream is the recorded sequence of clicks a visitor makes on a website, representing their navigation path.

QUESTION 141

What is the significance of a data warehouse bus matrix?

Answer

The data warehouse bus matrix serves as a planning and communication tool, helping to visualize relationships between data marts and common dimensions. It aids in prioritizing which dimensions to standardize for effective data architecture.

QUESTION 142

How is the reserve established in the claims process?

Answer

A reserve is a preliminary estimate of the insurance company's potential liability for a claim, established when a claim is opened.

QUESTION 143

What is a factless fact table?

Answer

A factless fact table captures relationships between keys without measurement metrics. It records events like product promotions without directly linking to sales data.

QUESTION 144

What type of analysis can be performed using the accumulating snapshot for admissions?

Answer

Analysis of applicant flow through the admissions process, identification of bottlenecks, and adjustments to recruitment strategies based on applicant characteristics.

Flashcards 145 - 156

QUESTION 145

Define Slowly Changing Dimensions (SCD).

Answer

The tendency of dimension rows to change gradually over time, categorized into types 1, 2, and 3 based on how changes are handled.

QUESTION 146

What is a common issue when bypassing the grain declaration step?

Answer

Bypassing the grain declaration can lead to confusion and inconsistencies in data interpretation, complicating the design process.

QUESTION 147

What is the impact of organizational culture on the requirements gathering process?

Answer

Organizational culture influences how requirements are gathered, the types of forums used, and the overall effectiveness of user engagement in the process.

QUESTION 148

What does a fact table represent in a dimensional model?

Answer

A fact table represents business performance measurements, typically numeric and additive, that are characterized by a composite key drawn from dimension tables.

QUESTION 149

What is a data mart?

Answer

A data mart is a subset of a data warehouse focused on a specific business area, containing a suite of associated fact tables that describe a particular business process.

QUESTION 150

What is a billing organization's primary objective in health care?

Answer

To prepare and send bills to payers, track payments, and analyze billing data across various dimensions like patient and provider.

QUESTION 151

What challenges arise with consolidated fact tables?

Answer

Consolidated fact tables can represent a dimensionality compromise and may not meet user requirements for more granular data.

QUESTION 152

What is the trend of managing by the numbers in organizations?

Answer

The trend emphasizes data-driven decision-making, where managers utilize key performance indicators (KPIs) to analyze operations, leading to improved efficiency and competitive advantage.

QUESTION 153

Describe the role of bridge tables in dimensional modeling.

Answer

Bridge tables are used to connect many-to-many relationships between dimensions, such as linking multiple customers to a single account without violating the granularity of the fact table.

QUESTION 154

What factors are included in the 'class of service' dimension?

Answer

The class of service dimension includes attributes indicating whether a passenger is in first class, business, or coach, and can help analyze customer preferences.

QUESTION 155

What is a 'cache' in web browsers?

Answer

A cache is a designated disk space in a web browser that stores temporary copies of web objects to speed up subsequent requests for the same content.

QUESTION 156

What is the significance of slowly changing dimensions (SCD) in data warehousing?

Answer

SCDs track changes in dimension attributes over time, allowing businesses to maintain historical accuracy in reporting and analytics.

Flashcards 157 - 168

QUESTION 157

What are some examples of demographic attributes that might be included in a minidimension?

Answer

Examples of demographic attributes include age, gender, income level, and number of children, which can be analyzed for customer segmentation.

QUESTION 158

What is Gross Margin Percent?

Answer

The gross profit expressed as a percentage of gross revenue.

QUESTION 159

How does the insurance industry use data analytics?

Answer

Data analytics is used to identify trends, track performance, assess risks, and enhance customer service through better information utilization.

QUESTION 160

What role does the Customer Dimension play in a dimensional model?

Answer

The Customer Dimension contains descriptive attributes about customers, such as Customer ID, Name, City, State, and more, which are used to slice and analyze the data in fact tables.

QUESTION 161

How does CRM contribute to business success?

Answer

By providing a comprehensive understanding of customer interactions, CRM can lead to increased sales effectiveness, improved customer satisfaction, and higher customer retention rates.

QUESTION 162

What are some common attributes found in a customer dimension?

Answer

Common attributes include name, address, demographic information, segmentation classifications, and scores related to customer behavior.

QUESTION 163

What is the grain of a dimensional schema?

Answer

The grain refers to the level of detail in a fact table, such as daily inventory by product at each store, which is essential for effective data analysis.

QUESTION 164

What are common attributes found in customer dimension tables?

Answer

Common attributes include customer name, address, segmentation attributes, dates of interaction, and aggregated facts related to customer behavior.

QUESTION 165

How does the privacy movement influence data warehouse architecture?

Answer

The privacy movement demands that data warehouses centralize and secure personal information, enforce strict access policies, and maintain audit trails for data usage.

QUESTION 166

What are multivalued dimensions in health care data modeling?

Answer

Multivalued dimensions allow for tracking multiple attributes, such as diagnoses, along with their effective dates and weighting factors for allocation.

QUESTION 167

What is the significance of the 'Rate Plan Code' in telecommunications?

Answer

The Rate Plan Code identifies the pricing model associated with a customer's service line, influencing billing and service usage metrics.

QUESTION 168

What are dimension attributes?

Answer

Dimension attributes are descriptive characteristics of the data in dimension tables, used to provide context to the numerical facts in fact tables.

Flashcards 169 - 180

QUESTION 169

Describe the role of the finance department in data management.

Answer

The finance department should focus on information dissemination and value-added analysis rather than simply report creation, enhancing overall financial insight.

QUESTION 170

Describe the publishing metaphor in the context of data warehousing.

Answer

The publishing metaphor compares the role of a data warehouse manager to that of a magazine editor, emphasizing the importance of understanding user needs and ensuring high-quality, relevant data is presented.

QUESTION 171

Explain Time Shifting of a Promotion.

Answer

The tendency for promotions to cause customers to delay purchases until the promotion and then refrain from purchasing afterwards for a period.

QUESTION 172

What is a Helper Table in the context of databases?

Answer

Also known as a Bridge Table, it is used to manage many-to-many relationships between data.

QUESTION 173

What is the challenge with large customer dimensions?

Answer

Large customer dimensions present challenges in browsing performance and change tracking due to the sheer volume of data and the frequency of changes.

QUESTION 174

Explain the role of a Date Key in a Date Dimension.

Answer

The Date Key serves as a primary identifier for each date entry, linking it to corresponding fact tables for time-based analysis.

QUESTION 175

What is the difference between additive and non-additive facts?

Answer

Additive facts can be summed across dimensions, while non-additive facts, like percentages or ratios, cannot be directly summed.

QUESTION 176

What is the role of procurement in supply chain management?

Answer

Procurement involves acquiring goods and services needed by an organization, focusing on obtaining the best value while ensuring timely delivery.

QUESTION 177

What does 'ETL' stand for, and what is its purpose?

Answer

ETL stands for Extract, Transform, Load; it refers to the processes used to gather, process, and load data into the data warehouse.

QUESTION 178

What is the significance of the referrer in clickstream data?

Answer

The referrer provides information about the URL of the page that linked to the current request, allowing for tracking of visitor behavior and advertising effectiveness.

QUESTION 179

What is the process of allocating in data warehousing?

Answer

Allocating refers to the procedure of distributing parent order facts to the child line-item level, allowing for analysis across all order facts by various dimensions like product.

QUESTION 180

What is dimensional modeling?

Answer

Dimensional modeling is a design technique used in data warehousing that structures data into fact and dimension tables to make it intuitive and easy for users to query.

Flashcards 181 - 192

QUESTION 181

List some dimensions that are important in health care billing.

Answer

Important dimensions include patient, provider, diagnosis, treatment, date, and payer.

QUESTION 182

How are new surrogate keys assigned to unmatched operational keys?

Answer

New surrogate keys are assigned for unmatched operational keys with a descriptive attribute to identify the unknown operational key, ensuring accurate tracking and future updates.

QUESTION 183

What are the advantages of maintaining a simple, flat dimension table?

Answer

Maintaining a simple, flat dimension table allows for easier querying and better performance, as complex joins and navigational issues are minimized.

QUESTION 184

What does the term 'churn' mean in customer management?

Answer

Churn refers to the ratio of customers lost to customers gained within a subscription service, indicating customer retention and satisfaction.

QUESTION 185

Why is user input vital in dimensional modeling?

Answer

User input ensures that the dimensional model aligns with actual business requirements and user expectations, leading to more effective data analysis.

QUESTION 186

What are heterogeneous products in financial services?

Answer

Heterogeneous products are financial products that possess unique attributes and metrics, requiring specialized handling in a data warehouse to manage their complexity.

QUESTION 187

Describe the importance of documentation in dimensional modeling.

Answer

Documentation helps maintain clarity, enables communication among stakeholders, and supports ongoing maintenance and updates of the data model.

QUESTION 188

Why are dimension tables important?

Answer

Dimension tables provide textual descriptors that enhance the usability and understanding of the data warehouse, allowing for meaningful queries and analysis.

QUESTION 189

What is the purpose of views in the context of date dimensions?

Answer

Views allow multiple representations of the same date dimension to be joined separately to a fact table, accommodating different date roles without requiring multiple tables.

QUESTION 190

How do dimension authorities manage the publication of dimensions?

Answer

Dimension authorities manage the publication by ensuring that all downstream data marts receive consistent and updated dimensions simultaneously.

QUESTION 191

What techniques can be used for managing historical assignments of sales reps to customers?

Answer

Creating a factless fact table can track historical assignments even without sales activity, allowing for comprehensive analysis of sales rep coverage over time.

QUESTION 192

What are the components of the technical architecture in data warehousing?

Answer

The technical architecture includes data staging services, data access services, and metadata components to support integration and communication within the data warehouse.

Flashcards 193 - 204

QUESTION 193

What is the purpose of the transaction sequence number?

Answer

The transaction sequence number ensures the reliable reconstruction of transaction order within a fact table, important for maintaining accurate account balances.

QUESTION 194

Describe the relationship between data warehousing and customer relationship management.

Answer

Data warehousing provides the centralized data needed for effective customer relationship management, allowing businesses to analyze customer behavior and improve service.

QUESTION 195

What is a 'frequent flyer profile minidimension'?

Answer

A frequent flyer profile minidimension captures unique combinations of frequent flyer tier, home airport, and club membership status, enhancing analysis of customer loyalty.

QUESTION 196

Describe the term 'ETL' in data warehousing.

Answer

ETL stands for Extract-Transform-Load, referring to the processes that prepare operational data for the data warehouse by extracting, transforming, and loading it.

QUESTION 197

What is 'cannibalization' in the context of sales?

Answer

Cannibalization occurs when the growth of sales of one product leads to a decrease in sales of another product, typically within the same brand or manufacturer.

QUESTION 198

What are slowly changing dimensions in the context of procurement?

Answer

Slowly changing dimensions refer to descriptive attributes in dimension tables that may change over time, such as product hierarchies or vendor details, necessitating specific techniques for managing these changes.

QUESTION 199

What is the importance of comprehensive attribute definitions in a data warehouse?

Answer

Comprehensive attribute definitions ensure that users can effectively explore and analyze data, leading to better decision-making based on accurate insights.

QUESTION 200

Why is conformed dimension integration important in data warehouses?

Answer

Conformed dimensions ensure consistency across different analytic solutions, allowing for comprehensive analysis and avoiding data silos.

QUESTION 201

What are the steps involved in designing a dimensional model?

Answer

Steps include requirements gathering, drafting a bus matrix, defining dimensions and facts, and implementing the model in a data warehouse.

QUESTION 202

What should be prioritized during the specification of analytic applications?

Answer

The initial focus should be on the most critical capabilities needed by users, ensuring manageable expectations and timely delivery.

QUESTION 203

What is the significance of cash flow management in relation to commitments and payments?

Answer

The finance department monitors the relationship between commitments and payments to manage the organization's cash flow effectively.

QUESTION 204

What is the budgeting process in accounting?

Answer

The budgeting process involves planning and controlling financial resources based on projected revenues and expenses, including the creation of budgets and tracking performance against them.

Flashcards 205 - 216

QUESTION 205

How do business requirements influence data warehouse design?

Answer

Business requirements must be clearly understood and translated into design considerations. They impact nearly every decision made during warehouse development.

QUESTION 206

What is the insurance value chain?

Answer

The insurance value chain refers to the series of steps involved in the insurance process, from policy creation and underwriting to claims processing and customer service.

QUESTION 207

What are 'degenerate dimensions'?

Answer

Degenerate dimensions are fact table attributes that do not have their own dimension table, often including transaction numbers.

QUESTION 208

What are the common pitfalls of data warehousing?

Answer

Common pitfalls include focusing too much on technology instead of user needs, creating overly complex schemas, and failing to ensure data quality and consistency.

QUESTION 209

What is an example of a causal dimension in retail sales?

Answer

Causal dimensions might include ads, promotions, or discounts that are believed to influence sales performance.

QUESTION 210

What challenges arise from packaged applications in data warehousing?

Answer

Packaged applications can limit control over data management due to proprietary systems, necessitating careful integration with the data warehouse for effective data extraction.

QUESTION 211

Define the term 'impact report' in health care billing.

Answer

An impact report summarizes billing amounts based on diagnoses or treatments without necessarily applying weighting factors.

QUESTION 212

What is clickstream data?

Answer

Clickstream data records the sequence of clicks a user makes on a website, providing insights into user behavior and website performance for e-commerce analysis.

QUESTION 213

What are personal data-collecting transducers?

Answer

Personal data-collecting transducers are devices that track and record individual movements and actions, contributing to data collection for various applications.

QUESTION 214

What is procurement in business?

Answer

Procurement is the process of acquiring goods and services from suppliers, encompassing the identification of needs, sourcing, negotiation, and purchasing.

QUESTION 215

Explain the concept of a minidimension in dimensional modeling.

Answer

A minidimension is a smaller dimension, often created for frequently changing attributes, that allows more efficient querying and analysis in large dimensional models.

QUESTION 216

Explain the concept of conformed dimensions.

Answer

Dimensions are conformed when they are either exactly the same or one is a perfect subset of the other, allowing for consistent interpretation across different fact tables.

Flashcards 217 - 228

QUESTION 217

What are conformed dimensions?

Answer

Conformed dimensions are dimensions that are identical or carefully chosen subsets across multiple fact tables, allowing for consistent data querying and reporting across different data marts.

QUESTION 218

What are some common attributes of the customer dimension in CRM?

Answer

Common attributes include customer ID, name, address, segmentation attributes, and dates relevant to customer interactions, facilitating detailed customer analysis.

QUESTION 219

What is the grain level chosen for the frequent flyer analysis?

Answer

The grain level chosen for frequent flyer analysis is at the segment level, allowing detailed insights into flight activity for marketing purposes.

QUESTION 220

What are some common indexing strategies for fact tables?

Answer

Common strategies include creating a concatenated index on primary dimensions, placing indexes on date keys, and optimizing for query performance.

QUESTION 221

How does the Web profitability data mart extend the sales transaction data mart?

Answer

It allocates activity and infrastructure costs to each sales transaction to provide a comprehensive view of profitability across all sales channels.

QUESTION 222

What is the accumulating snapshot grain in the context of health care billing?

Answer

The accumulating snapshot grain captures the history of a billing line item from its creation to the current day, allowing for tracking of changes over time.

QUESTION 223

What is the role of the Chief Information Officer (CIO)?

Answer

The CIO is a senior executive responsible for managing and directing an organization's IT strategy and infrastructure, ensuring alignment with business objectives.

QUESTION 224

Define Impact Report.

Answer

A report where the weighting factor of a multivalued dimension is ignored, providing summary totals regardless of the extent of involvement.

QUESTION 225

What are the challenges associated with the patient and responsible party dimensions in health care?

Answer

Challenges include the lack of a reliable national identity number and the difficulty of consolidating patient records across different providers.

QUESTION 226

What is meant by 'claims processing efficiency'?

Answer

Claims processing efficiency refers to the effectiveness and speed with which claims are handled, often measured by the time lag between the claim open date and the first payment date.

QUESTION 227

What is meant by 'conforming dimensions'?

Answer

Conforming dimensions means ensuring that dimensions shared between fact tables are identical or carefully chosen subsets, allowing for accurate cross-source analysis.

QUESTION 228

How can bridge tables enhance the analysis of hierarchical customer structures?

Answer

Bridge tables enhance analysis by allowing for the representation of complex hierarchical relationships without altering the existing customer or fact tables.

Flashcards 229 - 240

QUESTION 229

What is the significance of the 'Geographic Location Dimension' in telecommunications?

Answer

The Geographic Location Dimension provides precise geographic attributes that are essential for mapping and analyzing data related to service lines, billing, and customer locations.

QUESTION 230

What are the types of slowly changing dimensions?

Answer

There are three primary types of slowly changing dimensions: Type 1 (overwrite the old value), Type 2 (create a new row for historical data), and Type 3 (add a new column for the new value).

QUESTION 231

What is the significance of cross-functional teams in data warehouse projects?

Answer

Cross-functional teams ensure that diverse perspectives and expertise are incorporated into the data warehouse design, enhancing its relevance and usability across the organization.

QUESTION 232

What are the possible types of fact tables in order management?

Answer

The three types of fact tables are transaction fact tables, periodic snapshot fact tables, and accumulating snapshot fact tables.

QUESTION 233

What is the purpose of a cyclic redundancy checksum (CRC) in dimensional modeling?

Answer

A CRC is used to quickly determine if rows in a master dimension cross-reference table have changed. If the CRC values match, the extracted record can be ignored; if they differ, each column must be checked to assess changes.

QUESTION 234

What is the purpose of the data warehouse bus matrix?

Answer

The bus matrix serves as a planning and communication tool, documenting business processes and their relationships to common dimensions across the data warehouse.

QUESTION 235

How can a Factless Fact Table be used in a university context?

Answer

It can track student registrations or events without measurable facts, focusing on the relationships between students, courses, and terms.

QUESTION 236

Define business dimensional lifecycle.

Answer

The business dimensional lifecycle is a methodology for planning, designing, implementing, and maintaining data warehouses, as outlined in The Data Warehouse Lifecycle Toolkit.

QUESTION 237

Why are minidimensions useful in financial services?

Answer

Minidimensions allow financial organizations to efficiently track changes in attributes, such as monthly credit bureau data, without relying on type 2 SCD techniques that can be inefficient with large dimensions.

QUESTION 238

How can the complexity of heterogeneous products be managed?

Answer

The complexity of heterogeneous products can be managed through a well-designed schema that separates core and custom attributes while maintaining clear relationships.

QUESTION 239

Why is it important to have readable text strings in a product dimension table?

Answer

Readable text strings improve the user experience, as businesspeople may not be familiar with cryptic codes. Clear labels help ensure accurate query constraints and report generation.

QUESTION 240

How can new dimensions be added to a dimensional model?

Answer

New dimensions can be added by creating a new dimension table and linking it to the fact table with a foreign key.

Flashcards 241 - 252

QUESTION 241

What considerations should be taken for customer geographic attributes?

Answer

Considerations include universal representation, cultural correctness, and differences in addresses for both domestic and international customers.

QUESTION 242

What is the purpose of a coverage table in the context of facilities utilization?

Answer

The coverage table helps track the usage of facilities over time, documenting which facilities were used for what purpose during each hour of the day.

QUESTION 243

What are some key attributes of the customer ship-to dimension?

Answer

Attributes include Customer Ship To ID, Name, Address, City, State, Zip Code, and Customer Credit Rating.

QUESTION 244

What is a minidimension?

Answer

A minidimension is a separate dimension created to handle frequently analyzed or changing attributes from a larger dimension, allowing for better performance and change tracking.

QUESTION 245

What is a blended transaction fact table?

Answer

A blended transaction fact table combines multiple transaction types into a single table, allowing users to analyze all procurement transactions together.

QUESTION 246

What are some attributes tracked in financial services?

Answer

Attributes such as monthly credit bureau scores, demographic data, retention, profitability, and delinquency characteristics are commonly tracked.

QUESTION 247

What is the significance of claims transactions in insurance?

Answer

Claims transactions are vital as they represent the costs associated with insurance policies, including the process of opening, adjusting, and closing claims.

QUESTION 248

What is the importance of quality assurance in the fact table data staging process?

Answer

Quality assurance is essential to ensure that the fact table data matches the extract statistics and meets the expected standards for accuracy and reliability.

QUESTION 249

What are syndicated data suppliers?

Answer

Syndicated data suppliers are companies that collect, clean, package, and resell data. Key suppliers include A.C. Nielsen and IRI for grocery data, and IMS Health for pharmaceutical data.

QUESTION 250

What challenges are associated with allocating costs to web profitability?

Answer

Allocating costs to web profitability can be challenging due to the difficulty in associating certain costs with specific sessions or transactions, especially when there's no immediate sale.

QUESTION 251

What are some interesting questions that can be answered using the student attendance event fact table?

Answer

Questions include which courses had the highest attendance, which students attended specific courses, and which faculty members taught the most students.

QUESTION 252

Explain the concept of Normalization in databases.

Answer

A modeling technique that removes redundancy by separating data into discrete entities, each becoming a table in a relational DBMS.

Flashcards 253 - 264

QUESTION 253

What is a reserve in insurance claims?

Answer

A reserve is a preliminary estimate of the insurance company's eventual liability for a claim, established when a new claim is opened. It can be adjusted as more information becomes available.

QUESTION 254

What is a Type 3 slowly changing dimension?

Answer

A Type 3 slowly changing dimension adds an additional column to capture the previous value of an attribute, allowing analysis of data as if the change never occurred.

QUESTION 255

What are the steps involved in a web client-server interaction?

Answer

The steps include clicking a link, sending an HTTP request, retrieving the requested document, and potentially making additional requests for other resources.

QUESTION 256

Explain how multiple currencies are handled in the G/L.

Answer

The G/L can store financial data in both local and standardized corporate currencies, allowing for accurate financial reporting and analysis.

QUESTION 257

What is the significance of a trial period during product selection?

Answer

A trial period allows organizations to test the product in their environment before making a full commitment, ensuring it meets their needs.

QUESTION 258

What is the significance of interviewing data sources and subject matter experts?

Answer

These interviews assess the feasibility of supporting business needs with existing data and ensure that the necessary core data exists before proceeding.

QUESTION 259

What are the challenges with relying solely on SQL date semantics in querying?

Answer

SQL date functions do not support nonstandard attributes such as fiscal periods, weekends, and holidays, making it essential to use a date dimension for effective analysis.

QUESTION 260

Explain the importance of having separate fact tables for different procurement processes.

Answer

Separate fact tables allow for richer, more descriptive dimensions and attributes tailored to each distinct business process, avoiding generalization.

QUESTION 261

What is the significance of integration in data warehousing?

Answer

Integration is crucial for consistency in reporting and analysis across different data sources, enabling a comprehensive view of organizational performance.

QUESTION 262

What is a dimension in the context of data warehousing?

Answer

A dimension is a structure that categorizes facts and measures in order to enable users to answer business questions. It is often used in conjunction with fact tables in a star or snowflake schema.

QUESTION 263

What is the difference between blended and separate transaction schemas?

Answer

A blended transaction schema combines multiple types of transactions into a single fact table, while separate transaction schemas create distinct fact tables for each transaction type, reflecting unique business processes.

QUESTION 264

What does it mean when a dimension is snowflaked?

Answer

A dimension is snowflaked when low-cardinality columns are removed to separate normalized tables that link back to the original dimension table, increasing complexity and reducing browsing performance.

Flashcards 265 - 276

QUESTION 265

What are some expected applications of the Internet appliance?

Answer

Applications include information search and retrieval, electronic mail, telephony, mobile teleconferencing, e-learning, shopping, navigation, and language translation.

QUESTION 266

What is the role of packet-filtering gateways in data warehouses?

Answer

Packet-filtering gateways isolate key servers, allowing them to only receive packets from trusted application servers, thereby enhancing security and limiting exposure to potential attacks.

QUESTION 267

What are the implications of having semiadditive inventory facts?

Answer

It complicates calculations like averages over time, as SQL functions may not correctly aggregate data across the date dimension.

QUESTION 268

What are the key steps in the data warehouse lifecycle?

Answer

The key steps include project planning, defining business requirements, dimensional modeling, physical design, data staging, and deployment.

QUESTION 269

Explain the concept of a star join schema.

Answer

A star join schema is a database design where a central fact table is connected to several dimension tables, forming a star-like structure that simplifies queries.

QUESTION 270

What is a composite key?

Answer

A composite key is a combination of two or more columns in a database table that can uniquely identify a row in that table.

QUESTION 271

Why should calculated facts be stored physically in the database?

Answer

Storing calculated facts like gross profit ensures consistency and reduces the risk of user error in reporting, despite minor storage costs.

QUESTION 272

What is snowflaking in dimensional modeling?

Answer

Snowflaking is a normalization process where dimension tables are split into additional tables, creating a more complex structure compared to denormalized dimension tables.

QUESTION 273

How can promotions impact product sales in a grocery store?

Answer

Promotions can lead to significant increases in sales volume, particularly when combined with effective marketing strategies.

QUESTION 274

What does the term 'data warehouse' refer to?

Answer

A data warehouse is a central repository for integrated data from multiple sources, designed to support business intelligence activities, including analytics and reporting.

QUESTION 275

What is a student dimension in the context of student registration events?

Answer

The student dimension includes attributes related to students, such as their ID, status (part-time/full-time), and declared major.

QUESTION 276

What is the purpose of an explicit date dimension table in a data warehouse?

Answer

An explicit date dimension table organizes nonstandard date attributes like holidays, fiscal periods, and seasons, allowing for easier querying and analysis without relying on complex SQL date functions.

Flashcards 277 - 288

QUESTION 277

How does customer data consolidation affect CRM?

Answer

Consolidation ensures unique customer identifiers and integrates disparate sources of customer information, aiding in data accuracy and reducing redundancy in customer communication.

QUESTION 278

What are facts in dimensional modeling?

Answer

Facts are the numeric measurements that users want to analyze, such as sales quantity, sales dollar amount, and cost dollar amount.

QUESTION 279

What is the relationship between facts and the grain in a dimensional model?

Answer

Facts must adhere to the grain defined for the fact table, ensuring that all measurements are consistent with the level of detail specified.

QUESTION 280

What does the term 'segment' refer to in the context of air travel?

Answer

A segment refers to a portion of a trip on a single aircraft, which may consist of one or more legs, and is critical for analyzing customer flight activity.

QUESTION 281

What is a conformed dimension in a data mart?

Answer

A conformed dimension is a dimension that is shared across multiple data marts, ensuring consistency in the data definitions and allowing for integrated analysis.

QUESTION 282

When should you use a Type 3 response?

Answer

Use a Type 3 response when there is a need to track both old and new values of an attribute simultaneously, especially in cases of reorganizations or label changes.

QUESTION 283

Define 'clickstream' in the context of data warehousing.

Answer

Clickstream refers to the recorded actions of a user on a web browser, capturing their navigation paths, which can be analyzed for user behavior insights.

QUESTION 284

Describe the Type 2 response to slowly changing dimensions.

Answer

The Type 2 response creates a new dimension row for each change, preserving the history of previous attribute values in the fact table.

QUESTION 285

What are year-to-date facts in financial data?

Answer

Year-to-date facts are cumulative totals that should not be stored directly in the fact table due to potential inaccuracies; they should be calculated instead.

QUESTION 286

What is a glossary in the context of data warehousing?

Answer

A glossary provides definitions of key terms and concepts in data warehousing, serving as a reference for users to understand the terminology used in the field.

QUESTION 287

What is a multivalued dimension, and how is it handled in health care billing?

Answer

A multivalued dimension allows for multiple values in the context of a single measurement, such as multiple diagnoses. It is managed through a bridge table that connects the fact table to the multiple values.

QUESTION 288

What is Market Basket Analysis?

Answer

An analysis in retail that seeks to understand all products purchased by a customer in a single shopping event, often used for affinity grouping.

Flashcards 289 - 300

QUESTION 289

What role does education play in shaping data warehousing practices?

Answer

Education helps stakeholders understand the importance of data management, privacy concerns, and the potential of data warehousing to drive business success.

QUESTION 290

What is 'gross profit'?

Answer

Gross profit is the total revenue minus the cost of goods sold, representing the profit a company makes from its core business activities before operating expenses.

QUESTION 291

What does HTTP stand for?

Answer

HyperText Transfer Protocol, which specifies how a browser and web server exchange information.

QUESTION 292

What is Household in marketing terms?

Answer

The effort to assign an account or individual to a household of accounts or individuals for marketing purposes.

QUESTION 293

How does a behavior study group function?

Answer

A behavior study group captures the keys of a specific set of customers identified through complex queries for further analysis without reprocessing to identify the initial conditions.

QUESTION 294

What is an accumulating snapshot fact table?

Answer

An accumulating snapshot fact table records multiple dates representing major milestones in a process, updated as activities occur, providing a timeline of events.

QUESTION 295

What is Gross Revenue?

Answer

The total revenue paid to a company by its customers. If calculated before discounts, the actual amount paid is called net revenue.

QUESTION 296

What are the implications of having a national ID system on data warehousing?

Answer

A national ID system could streamline data management and enhance security but also raises concerns about privacy and the potential misuse of personal information.

QUESTION 297

What is the significance of understanding the source data?

Answer

Understanding the source data is essential for accurately defining dimensions and facts that accurately reflect the business processes.

QUESTION 298

What is meant by 'dimension outriggers' in data modeling?

Answer

Dimension outriggers are additional tables used to store low-cardinality attributes that can be linked to a primary dimension table, allowing for better organization and retrieval of data.

QUESTION 299

Describe the role of a data warehouse bus architecture.

Answer

The data warehouse bus architecture is a framework that organizes the presentation area of the data warehouse based on conformed dimensions and facts, allowing for efficient data integration.

QUESTION 300

What is the potential downside of presenting users with multiple avenues for accessing demographic data?

Answer

Presenting users with multiple avenues can lead to confusion and complexity, making it harder for them to know which route to take for their analysis.

Flashcards 301 - 312

QUESTION 301

What responsibilities does an underwriter have in the insurance policy process?

Answer

An underwriter is responsible for making the final approval and taking ultimate responsibility for business dealings with the policyholder.

QUESTION 302

What is the challenge of over-normalizing customer dimensions?

Answer

Over-normalizing customer dimensions can lead to complex designs that are difficult for users to navigate, potentially hindering performance and usability.

QUESTION 303

Why is user involvement critical in a data warehouse project?

Answer

User involvement is crucial for acceptance and success; it ensures that the project meets actual needs and fosters a sense of ownership among users.

QUESTION 304

What is the significance of having a detailed project plan?

Answer

A detailed project plan outlines responsibilities, timelines, and deliverables, ensuring that all team members are aligned and that the project stays on track.

QUESTION 305

What is the role of audit databases in data warehousing?

Answer

Audit databases track the use of personal information, maintaining records of who accessed the data and for what purpose, enhancing accountability and transparency.

QUESTION 306

Why is the design of dimension attributes critical in a data warehouse?

Answer

The design of dimension attributes is critical because they are key to making the data warehouse usable and understandable, influencing the quality of queries and reports generated.

QUESTION 307

Explain the importance of tracking changes to policyholder dimensions.

Answer

Tracking changes to policyholder dimensions is crucial for accurate pricing and risk assessment, especially for attributes like ZIP codes that impact underwriting.

QUESTION 308

What is meant by 'degenerate dimensions'?

Answer

Degenerate dimensions refer to control numbers created at various steps in the procurement process that do not have their own dimension but are still important for analysis.

QUESTION 309

What is the importance of a project launch meeting?

Answer

A project launch meeting emphasizes the business sponsor's commitment, helps generate a sense of ownership among participants, and disseminates a consistent project message.

QUESTION 310

How are keywords used in the HR dimension for skills?

Answer

Keywords describe employee skills and are managed as multivalued attributes that allow querying based on various technical proficiencies.

QUESTION 311

What is a many-to-many relationship in dimensional modeling?

Answer

A relationship where multiple records in one dimension can relate to multiple records in another, necessitating separate dimensions with their own foreign keys.

QUESTION 312

What are key facts tracked in an accumulating snapshot for health care billing?

Answer

Key facts include billed amount, amounts paid by primary and secondary insurance, remaining balance, and payment timelines.

Flashcards 313 - 324

QUESTION 313

Explain the concept of a multivalued dimension.

Answer

A multivalued dimension allows for the association of multiple attributes to a single entry in a dimension table, useful for capturing data such as skills that vary in number across entries.

QUESTION 314

What are the key components of warehouse inventory transaction modeling?

Answer

Key components include date, product, warehouse, vendor, transaction type, and inventory quantity impact, providing detailed insights into inventory movements.

QUESTION 315

What are the challenges of combining core and custom schemas?

Answer

Combining core and custom schemas can lead to complexity and potential data holes if not managed properly; typically, these schemas are kept separate.

QUESTION 316

What are some methods to ensure accurate clickstream data collection despite caching?

Answer

Using 'No Cache' HTML tags or setting expiration dates can help ensure that browsers retrieve the most current content from the server.

QUESTION 317

What is the first step in the dimensional design process?

Answer

The first step is to select the business process that will be modeled, combining business requirements with available data.

QUESTION 318

Define 'cross-selling' in retail.

Answer

Cross-selling is a technique used to increase sales by promoting new product lines to existing customers, thereby enhancing the overall sales volume.

QUESTION 319

What is Market Growth?

Answer

A desirable outcome of a promotion that increases overall sales of a product category rather than causing cannibalization.

QUESTION 320

What is a minidimension?

Answer

A minidimension is a smaller dimension created to handle frequently changing attributes that can be linked back to a larger fact table, allowing for efficient tracking of changes without bloating the main dimension.

QUESTION 321

What are the benefits of using a centralized approach to data warehousing?

Answer

A centralized approach helps ensure consistency and integrity across data sources, providing a single source of truth for reporting and analysis.

QUESTION 322

What is an example of a Junk Dimension?

Answer

A dimension that holds miscellaneous attributes that do not belong to any other dimension, often used to simplify data models.

QUESTION 323

What are proxy servers and how do they impact clickstream data?

Answer

Proxy servers cache content to reduce traffic but can deliver outdated content, fail to notify the originating server, and obscure the identity of visitors.

QUESTION 324

What metrics are often used to evaluate promotional success?

Answer

Metrics include sales lift during promotions, sales shifts before and after promotions, and overall profitability taking promotional costs into account.

Flashcards 325 - 336

QUESTION 325

How can GIS capabilities be integrated with a data warehouse?

Answer

GIS can be integrated by standardizing address information into a parsed form that can be matched with geographic points in a network database, enabling spatial analysis.

QUESTION 326

What is a core fact table?

Answer

A core fact table contains performance metrics that are common across all lines of business, allowing for comprehensive analysis across diverse products.

QUESTION 327

How can SQL performance be improved when handling large keyword dimensions?

Answer

Performance can be improved by pinning keyword lists in memory or creating specialized indexes that allow for faster searches without exhaustive table scans.

QUESTION 328

What are the main activities involved in the procurement process?

Answer

The procurement process includes negotiating contracts, issuing purchase requisitions and orders, tracking receipts, and authorizing payments, all of which have analytical requirements to monitor performance and compliance.

QUESTION 329

What are some key dimensions associated with health care billing?

Answer

Key dimensions include responsible party, employer, patient, provider, treatment location, diagnosis, and insurance organizations.

QUESTION 330

How do surrogate keys support the handling of changes to dimension table attributes?

Answer

Surrogate keys provide a mechanism to differentiate historical instances of the same operational account number, essential for managing slowly changing dimensions.

QUESTION 331

What challenges exist when maintaining an organization bridge table?

Answer

Challenges include identifying organizational changes, handling these changes without affecting internal paths, and managing historical data effectively.

QUESTION 332

What are arbitrary value bands in reporting?

Answer

Arbitrary value bands allow users to create flexible reporting that groups values into ranges, which can vary in size and definition based on user needs.

QUESTION 333

What does the term 'grain' refer to in the context of a fact table?

Answer

The grain specifies the level of detail contained in the fact table, for example, one row per individual transaction or event.

QUESTION 334

How does dimensional modeling apply across different industries?

Answer

Dimensional modeling techniques are transferable across industries, including banking, insurance, health care, and retail, as they address common data analysis challenges.

QUESTION 335

Why is it essential to manage expectations during the data warehousing process?

Answer

Managing expectations helps ensure that all stakeholders understand the project scope, limitations, and potential outcomes, reducing the risk of dissatisfaction later.

QUESTION 336

What is the importance of conducting detailed evaluations during product selection?

Answer

Detailed evaluations ensure that the selected product meets both functional and technical requirements, aligning with the architectural goals.

Flashcards 337 - 348

QUESTION 337

What types of procurement transactions are commonly analyzed?

Answer

Common procurement transactions include purchase requisitions, purchase orders, shipping notifications, receipts, and payments, each representing different processes within the procurement cycle.

QUESTION 338

What is the purpose of customer matching in CRM?

Answer

Customer matching aims to identify unique customer records across various systems, utilizing algorithms and reference files to validate and merge customer data effectively.

QUESTION 339

What is the importance of daily backups in data warehouses?

Answer

Daily backups safeguard against data loss by ensuring that critical information is securely stored and can be recovered in case of a system failure or catastrophic event.

QUESTION 340

What is the purpose of dimensional modeling in data warehousing?

Answer

Dimensional modeling simplifies the database design, making it intuitive for users to navigate and query data while enhancing performance.

QUESTION 341

How does CRM require organizational commitment?

Answer

CRM requires a commitment to integrate processes, technology, and people across the organization to effectively serve customer needs and foster loyalty.

QUESTION 342

What is a consolidated fact table?

Answer

A consolidated fact table combines metrics from separate business processes into a single table for easier analysis, typically at a common granularity.

QUESTION 343

What is the relationship between the retail sales schema and SQL queries?

Answer

The retail sales schema is designed to facilitate complex queries by using dimension constraints, leading to insightful results about sales data.

QUESTION 344

What does 'Webhouse' refer to?

Answer

A Webhouse is an evolved form of a data warehouse that adapts to the existence of the Web, integrating web-based data storage and management.

QUESTION 345

Explain 'Type 2' Slowly Changing Dimension (SCD).

Answer

Type 2 SCD involves creating a new dimension record with a new surrogate key to capture changes, preserving historical data.

QUESTION 346

How can a proof of concept be useful in a data warehouse project?

Answer

A proof of concept demonstrates potential capabilities, helping to align user expectations and garner support for the data warehouse initiative.

QUESTION 347

Explain the concept of explicit rows for nonattendance events.

Answer

Explicit rows can be added to track students who were registered but did not attend, allowing for analysis of attendance patterns.

QUESTION 348

What role do business representatives play in the evaluation process?

Answer

Business representatives provide insights on data access needs and help drive the evaluation process to ensure alignment with business objectives.

Flashcards 349 - 360

QUESTION 349

What is the purpose of a frequent shopper dimension?

Answer

The frequent shopper dimension allows businesses to analyze specific shopper behaviors and patterns by linking transactions to individual shoppers.

QUESTION 350

What does the Primary Insurance Billing Date Key represent?

Answer

The Primary Insurance Billing Date Key is a foreign key that indicates the date when the primary insurance was billed for services provided.

QUESTION 351

What does 'snowflaking' mean in the context of dimensional modeling?

Answer

Snowflaking refers to the normalization of dimension tables, where attributes are split into additional tables to reduce redundancy, though it can complicate query performance.

QUESTION 352

What is the purpose of conformed dimensions?

Answer

Conformed dimensions ensure consistency across fact tables and allow for integrated analysis across different business processes within the data warehouse.

QUESTION 353

What is a primary key in the context of dimensional modeling?

Answer

A primary key uniquely identifies each record in a dimension or fact table, ensuring data integrity.

QUESTION 354

What attributes might be included in an Applicant Dimension?

Answer

Attributes such as applicant name, GPA, SAT scores, gender, ethnicity, and preliminary major.

QUESTION 355

How is the applicant tracking process similar to an accumulating snapshot?

Answer

Both involve tracking an entity through various stages, updating the status and attributes as new information is gathered.

QUESTION 356

What is clickstream data?

Answer

Clickstream data records every page event from a web visitor, capturing information about user interactions on a website.

QUESTION 357

Why is it essential to create conformed dimensions in CRM?

Answer

Creating conformed dimensions ensures consistency and accuracy in customer data analysis, preventing the issues associated with disconnected data silos.

QUESTION 358

Define semiadditive facts in the context of inventory.

Answer

Semiadditive facts can be summed across some dimensions but not all; for example, inventory levels can be summarized by product or store but not over time, as they represent static snapshots.

QUESTION 359

What is the importance of capturing zero measurements in inventory?

Answer

Capturing zero measurements ensures comprehensive tracking of inventory levels, even for products that did not sell during a given period.

QUESTION 360

What are the benefits of using common dimensions in a data warehouse?

Answer

Common dimensions allow for integration across multiple data marts, facilitating consistent reporting and analysis across business processes.

Flashcards 361 - 372

QUESTION 361

How should a health care billing data warehouse be structured for analysis?

Answer

It should allow users to analyze billing data by various dimensions including patient, provider, treatment, and payment status, with simple and responsive queries.

QUESTION 362

Explain the concept of role-playing dimensions.

Answer

Role-playing dimensions are dimensions that can be used in different contexts within the same database, often implemented as views on a single physical table.

QUESTION 363

What is a merchandise hierarchy in the context of the product dimension?

Answer

A merchandise hierarchy organizes SKUs into brands, categories, and departments, allowing for structured analysis and reporting within retail data.

QUESTION 364

What dimensions are relevant in a clickstream environment?

Answer

Relevant dimensions include date, time, vendor, transaction status, product, customer, session, referral, and event.

QUESTION 365

What advantages does facilitated session gathering provide?

Answer

Facilitated sessions can reduce the time needed to gather requirements by encouraging group discussion and consensus-building among participants.

QUESTION 366

Define the grain of the policy transaction fact table.

Answer

The grain of the policy transaction fact table is one row for each individual policy transaction, capturing detailed context for each atomic transaction.

QUESTION 367

Describe the difference between inventory transactions and periodic snapshots.

Answer

Inventory transactions capture every event affecting inventory (e.g., receiving, shipping) and provide granular insights, while periodic snapshots summarize inventory levels at specified intervals.

QUESTION 368

How can procurement analytics help organizations?

Answer

Procurement analytics can identify frequently purchased materials, vendor performance, compliance with negotiated pricing, and opportunities for cost savings through supplier consolidation or improved purchasing behaviors.

QUESTION 369

What is an analytic application in the context of data warehousing?

Answer

Analytic applications are tools or templates designed to meet business users' analytical needs, enhancing the utility of the data warehouse.

QUESTION 370

What is the conformed customer dimension?

Answer

The conformed customer dimension is a well-maintained and standardized structure within a data warehouse that provides consistent customer information for analysis.

QUESTION 371

How can the profitability of a web sales channel be measured?

Answer

By allocating appropriate costs back to individual sales transactions, enabling a detailed analysis of profitability.

QUESTION 372

What are low-cardinality attributes?

Answer

Low-cardinality attributes are those that have a limited number of distinct values, making them suitable candidates for normalization and use in outriggers or minidimensions.

Flashcards 373 - 384

QUESTION 373

Describe the challenges of querying across multiple fact tables.

Answer

Querying across multiple fact tables can lead to incorrect results due to differences in cardinality and granularity, necessitating careful query design to ensure accurate data retrieval.

QUESTION 374

What is the significance of financial services in the economy?

Answer

Financial services facilitate the allocation of resources, provide liquidity, and support economic growth through various offerings like banking, investment, and insurance.

QUESTION 375

What is a procurement transaction date key?

Answer

A procurement transaction date key is a foreign key in the date dimension that links to the date of a specific procurement transaction.

QUESTION 376

What is the impact of web visitor behavior on profitability analysis?

Answer

Web visitor behavior can significantly influence profitability analysis by revealing patterns of engagement that correlate with sales, helping to allocate costs and predict revenue.

QUESTION 377

What is the grain of the profitability fact table?

Answer

The grain is defined as each individual product sold on a sales ticket to a customer at a specific point in time.

QUESTION 378

Define a 'Table' in database context.

Answer

A table is a collection of rows (records) that have associated columns (fields) within a database.

QUESTION 379

What is granularity in the context of a fact table?

Answer

Granularity refers to the level of detail captured in the data warehouse, specifically the meaning of a single row in a fact table.

QUESTION 380

How does the organizational roll-up function in the G/L?

Answer

The organizational roll-up links accounts to cost centers, departments, and divisions, providing a hierarchical view of financial data.

QUESTION 381

What is an Internet appliance?

Answer

An Internet appliance is a portable device that connects to the Internet, providing real-time audio, video, and communication capabilities, enhancing data collection and user interaction.

QUESTION 382

Explain what a composite key is.

Answer

A composite key is a database key made up of two or more columns, ensuring uniqueness in a table when a single column is insufficient.

QUESTION 383

What information is typically included in the promotion dimension?

Answer

The promotion dimension describes promotion conditions under which products are sold, including price reductions, advertising types, display types, and coupon types.

QUESTION 384

What is the relationship between policy transactions and profitability?

Answer

Policy transactions provide the detailed activity data necessary to calculate profitability across various dimensions.

Flashcards 385 - 396

QUESTION 385

What is the significance of the 'General Ledger' (G/L)?

Answer

The General Ledger (G/L) is a comprehensive accounting record that tracks an organization’s assets, liabilities, equity, income, and expenses, ensuring financial integrity.

QUESTION 386

What is the significance of a chart of accounts in accounting?

Answer

The chart of accounts is a list of accounts used by the general ledger, providing a uniform structure for tracking an organization's assets, liabilities, equity, income, and expenses.

QUESTION 387

What are Operational and Analytic CRM?

Answer

Operational CRM focuses on synchronizing customer-facing processes, while Analytic CRM leverages data analysis to optimize customer interactions and decision-making.

QUESTION 388

Explain the term 'fact dimension' in the context of medical records.

Answer

A fact dimension describes what each row in a medical record represents, accommodating the variability of data types and measurements.

QUESTION 389

What is a core product dimension table?

Answer

A core product dimension table includes attributes common across all products, serving as a reference for various product types within the data warehouse.

QUESTION 390

What are semi-additive facts in inventory models?

Answer

Semi-additive facts are measures that can be summed across some dimensions but not all, such as inventory levels, which can't be added across dates.

QUESTION 391

What is the role of data staging in the data warehouse architecture?

Answer

Data staging involves preparing and transforming data from source systems before it is loaded into the data warehouse, ensuring data quality and consistency.

QUESTION 392

What are first-level and second-level data marts?

Answer

First-level data marts are closely associated with single sources of data, while second-level data marts consolidate data from multiple first-level marts, such as profitability metrics.

QUESTION 393

What factors should be considered in designing a secure data warehouse?

Answer

Factors include data encryption, user authentication, regular security assessments, and compliance with regulatory requirements to protect sensitive information.

QUESTION 394

What does a Transaction Fact Table capture?

Answer

A Transaction Fact Table captures data at the lowest level of detail for each transaction event, with a record present only if an event occurs.

QUESTION 395

What is the role of the store dimension in a retail data mart?

Answer

The store dimension captures details about each store, including location attributes and characteristics, allowing for geographic analysis of sales data.

QUESTION 396

How can aggregate clickstream fact tables improve query performance?

Answer

Aggregate clickstream fact tables reduce data size by summarizing details into more manageable dimensions, thus significantly improving query performance by allowing faster access to summarized data.

Flashcards 397 - 408

QUESTION 397

How does the store dimension enable geographic analysis?

Answer

The store dimension allows stores to be rolled up into geographic attributes such as ZIP codes, counties, and regions, facilitating analysis of sales patterns based on location.

QUESTION 398

What is the consequence of premature summarization in a fact table?

Answer

Premature summarization limits the ability to add new dimensions that may not apply at the higher grain, complicating future data analysis.

QUESTION 399

What are the three inventory models discussed?

Answer

The three models are: periodic snapshot, transaction model, and accumulating snapshot model, each serving different analytical needs.

QUESTION 400

What are the goals of a data warehouse?

Answer

The goals of a data warehouse include providing a unified view of data, improving decision-making through data analysis, and enabling efficient data retrieval for business users.

QUESTION 401

What is the importance of maintaining data quality in a data warehouse?

Answer

Maintaining data quality ensures accurate reporting and analysis, as poor data quality can lead to misinterpretations and unreliable business insights.

QUESTION 402

Define the role of the store dimension in retail sales analysis.

Answer

The store dimension provides detailed attributes about each store, such as location and opening dates, which are critical for sales analysis.

QUESTION 403

What is the primary focus of dimensional modeling in a data warehouse?

Answer

Dimensional modeling focuses on designing data structures that allow for efficient and intuitive data retrieval and analysis, often using fact tables and dimension tables.

QUESTION 404

What complexities arise in health care data modeling?

Answer

Health care data modeling often deals with complex relationships among patients, treatments, and diagnoses, requiring advanced techniques for accurate representation.

QUESTION 405

In what ways can dimensional models be extended?

Answer

Dimensional models can be extended by adding new dimensions or facts, supplementing existing dimensions with new attributes, or breaking down existing rows to a lower granularity.

QUESTION 406

What are the challenges of using multivalued dimensions?

Answer

Multivalued dimensions can complicate queries, especially for AND conditions, requiring specific SQL strategies to handle constraints across rows.

QUESTION 407

What is a composite key in a fact table?

Answer

A composite key is formed from multiple foreign keys that uniquely identify each record in a fact table.

QUESTION 408

What is the main focus of Chapter 16?

Answer

Chapter 16 shifts focus from dimensional modeling techniques to the overall lifecycle of a data warehouse project, covering planning, design, and maintenance best practices.

Flashcards 409 - 420

QUESTION 409

What is a deal dimension in the context of order management?

Answer

The deal dimension describes the incentives and terms offered to customers that influence their purchasing decisions, including descriptions of deals, terms, allowances, and incentives.

QUESTION 410

What are micropayments, and how do they relate to e-wallets?

Answer

Micropayments are small transactions made for digital content, facilitated by secure e-wallets that authenticate users and allow for easy payment methods.

QUESTION 411

What are junk dimensions?

Answer

Junk dimensions are used to store miscellaneous attributes that do not fit neatly into other dimensions, helping to simplify fact tables and manage sparsely populated attributes.

QUESTION 412

Explain the role of aggregate navigators in data warehousing.

Answer

Aggregate navigators optimize queries by automatically directing requests to the most appropriate aggregate tables, enhancing performance.

QUESTION 413

What is an accumulating snapshot in the order fulfillment pipeline?

Answer

An accumulating snapshot is a fact table that updates existing rows as orders progress through various stages, providing an overview of order status and efficiency.

QUESTION 414

What is a star schema?

Answer

A star schema is a type of database schema that is characterized by a central fact table connected to multiple dimension tables, optimized for querying and reporting.

QUESTION 415

What are some challenges faced in integrating data in insurance?

Answer

Challenges include managing disparate data sources, ensuring data consistency, and overcoming historical organizational silos.

QUESTION 416

What is the significance of a bridge table in linking dimensions?

Answer

A bridge table is crucial for linking dimensions that have a many-to-many relationship, allowing for accurate representation of complex relationships in a fact table.

QUESTION 417

What are the primary goals of a data warehouse?

Answer

The primary goals of a data warehouse include providing a centralized repository for data analysis, improving query performance, and supporting decision-making processes.

QUESTION 418

What should the technical architecture design process focus on?

Answer

The design process should focus on high-value business needs, ensuring that technical solutions align with these requirements rather than just addressing technical challenges.

QUESTION 419

What is a consolidated fact table?

Answer

A consolidated fact table combines data from multiple business processes, allowing for comprehensive reporting on metrics such as premium revenue and claim losses.

QUESTION 420

What are the attributes of the product dimension?

Answer

Attributes include Vendor Key, Vendor Name, Vendor Address, Vendor City, Vendor Zip, Vendor State/Province, Vendor Country, Vendor Status, and Vendor Minority Ownership Flag.

Flashcards 421 - 432

QUESTION 421

What are consolidated data marts?

Answer

Consolidated data marts are complex data marts that integrate multiple data sources across different business processes to provide a comprehensive view, such as profitability. They are beneficial but more challenging to implement due to increased ETL efforts.

QUESTION 422

What is the foundation of dimensional design?

Answer

Dimensional design relies on data that is not aggregated to ensure users can drill down to atomic details. Normalized models may be useful for staging data but should not be used for presenting data to business users.

QUESTION 423

What are surrogate keys and why are they important?

Answer

Surrogate keys are unique identifiers used in data warehouses to ensure consistency and simplify data relationships.

QUESTION 424

What is the role of executive support in a data warehouse project?

Answer

Executive support is vital from the beginning to ensure that the data warehouse initiative is taken seriously and has the necessary resources and backing for successful implementation.

QUESTION 425

How can you prioritize data warehouse projects?

Answer

Projects can be prioritized using a four-cell quadrant matrix based on their potential impact and feasibility, focusing on high-impact, feasible projects.

QUESTION 426

What is the primary purpose of the data presentation area in a data warehouse?

Answer

The data presentation area organizes, stores, and makes data available for direct querying by users and analytical applications. It is essentially the interface through which the business community interacts with the data warehouse.

QUESTION 427

What are the sample values for the Referral Dimension attributes?

Answer

Sample values for the Referral Dimension include Referral Key, Referral Type (e.g., Intra Site, Remote Site), Referring URL, Referring Site, Referring Domain, and Search Type.

QUESTION 428

What is the purpose of the data warehouse bus matrix?

Answer

The data warehouse bus matrix helps to identify business processes and core data sources, facilitating a structured approach to dimensional modeling.

QUESTION 429

What is the benefit of using a snapshot in HR data?

Answer

Snapshots provide a quick view of employee statistics at month-end, allowing for summary reporting and trend analysis without detailed transaction review.

QUESTION 430

What is the contribution amount in a P&L statement?

Answer

The contribution amount represents the net income after deducting costs from the extended net invoice amount, indicating profitability before general expenses.

QUESTION 431

How do minidimensions improve performance?

Answer

Minidimensions improve performance by providing a smaller and more manageable set of attributes for analysis, allowing users to avoid querying large customer dimension tables directly.

QUESTION 432

What is the difference between header facts and line-item facts?

Answer

Header facts pertain to the overall order, while line-item facts detail individual items within the order, which allows for more granular analysis.

Flashcards 433 - 444

QUESTION 433

What is the role of a data modeler?

Answer

A data modeler creates the structural design of the data warehouse, focusing on how data is organized and accessed for analysis.

QUESTION 434

What is an accumulating snapshot in inventory management?

Answer

An accumulating snapshot tracks the status of inventory through various stages of processing in the warehouse, updating a single row as the product moves through defined milestones.

QUESTION 435

What are the main components of an HR periodic snapshot?

Answer

An HR periodic snapshot includes a fact table with dimensions for month, employee transaction, and organization, capturing monthly summaries.

QUESTION 436

How can data from multiple industries be interrelated through dimensional models?

Answer

Dimensional models can be designed to share common dimensions across industries, like customer and date dimensions, facilitating comprehensive cross-industry analyses.

QUESTION 437

What is the significance of customer status attributes in the Session Dimension?

Answer

Customer status attributes, such as 'high-value reliable customer' or 'in default', help in analyzing customer behavior over time and are derived from auxiliary data marts within the data Webhouse.

QUESTION 438

What is the purpose of using cookies in web analytics?

Answer

Cookies help identify and track individual web visitors, enabling better analysis of user behavior and session definitions.

QUESTION 439

What information does the HTTP request from a visitor's browser carry?

Answer

The HTTP request carries the referrer, which is the URL of the page that linked to the advertisement.

QUESTION 440

What is the purpose of data access tools in a data warehouse?

Answer

Data access tools allow users to query and analyze the data in the data warehouse's presentation area, facilitating decision-making and data-driven insights.

QUESTION 441

What challenges arise from the variability of medical records in data warehousing?

Answer

The variability makes it difficult to create a standardized framework for queries and reports, requiring flexible designs like fact dimensions.

QUESTION 442

What are the implications of having too many dimensions in a data warehouse?

Answer

Having too many dimensions can complicate the data model and hinder performance. Designers should aim for a balance between granularity and simplicity.

QUESTION 443

Why should dimension tables remain flat?

Answer

Flat dimension tables enhance usability and performance, enabling easier querying and browsing of attributes without complex joins.

QUESTION 444

Explain Hierarchical Relationship in data.

Answer

A relationship where data rolls up into higher levels of summarization in strict many-to-one relationships.

Flashcards 445 - 456

QUESTION 445

What challenges do organizations face when closing financial periods?

Answer

Organizations often face lengthy reconciliation processes and the need to ensure accuracy before officially closing financial results.

QUESTION 446

What is a Factless Fact Table?

Answer

A fact table that does not contain any measurable facts but instead captures events or the occurrences of relationships between dimensions.

QUESTION 447

Describe the role of the order number as a degenerate dimension.

Answer

The order number serves as a unique identifier for transactions, facilitating the grouping of order line items without requiring a separate dimension table.

QUESTION 448

What are some typical dimensions in a billing system tailored for hospital stays?

Answer

Typical dimensions may include admitting diagnosis, current diagnosis, discharge diagnosis, and different roles for providers involved in the patient's care.

QUESTION 449

What is the significance of the bus matrix in data mart design?

Answer

The bus matrix illustrates the dimensions used across different data marts, showing the relationships and shared dimensions to ensure integration and consistency in reporting.

QUESTION 450

What is a product evaluation matrix?

Answer

It is a spreadsheet that identifies evaluation criteria and weighting factors to assess potential products against the architecture plan.

QUESTION 451

How can performance be affected by value-band queries?

Answer

Value-band queries can be performance-intensive due to their lightly constrained nature, often requiring indexes on the balance facts to improve query speed.

QUESTION 452

Describe the relationship between multiple insured drivers and policyholders.

Answer

Multiple insured drivers can be associated with a policyholder, requiring a bridge table to capture their relationships and attributes for accurate premium calculations.

QUESTION 453

What challenges do people-related issues pose for CRM implementations?

Answer

People-related issues often involve resistance to change, as CRM requires new ways of interacting with customers and may involve significant alterations to organizational structures and incentive systems.

QUESTION 454

Why is user acceptance critical for a data warehouse?

Answer

User acceptance is critical because the success of a data warehouse hinges on its adoption by business users; if users do not embrace the system, its intended benefits will not be realized.

QUESTION 455

What is the purpose of a database management system (DBMS)?

Answer

A DBMS is a computer application designed to store, retrieve, and modify data in a structured way, facilitating data sharing among various applications.

QUESTION 456

What is the significance of understanding customer segments in insurance?

Answer

Understanding customer segments allows for tailored marketing strategies and better risk assessment, improving profitability.

Flashcards 457 - 468

QUESTION 457

What is the role of dimensions in a fact table?

Answer

Dimensions provide the contextual information necessary to interpret the numeric measurements in a fact table, allowing for detailed analysis and reporting.

QUESTION 458

What is the importance of having a predictable sequence in date dimensions?

Answer

A predictable sequence in date dimensions facilitates efficient partitioning and querying of fact tables, allowing for effective administrative actions on data.

QUESTION 459

What is the significance of the 'Appraised Value' in mortgage data?

Answer

Appraised Value refers to the estimated worth of a property as determined by a certified appraiser, which is critical for mortgage valuation and approval.

QUESTION 460

What are the different procurement transactions mentioned in the context?

Answer

The different procurement transactions include purchase orders, shipping notices, warehouse receipts, and vendor payments.

QUESTION 461

What are some analytical requirements that drive the use of bandwidth reporting?

Answer

Analytical requirements such as data mining and trend analysis often necessitate flexible reporting capabilities, which can be supported by banded attribute values.

QUESTION 462

What do high-quality dimension attributes provide?

Answer

High-quality dimension attributes serve as source constraints for queries and enhance the clarity of result sets.

QUESTION 463

What is the role of the date dimension in health care billing?

Answer

The date dimension helps track multiple date stamps associated with billing processes, allowing for accurate time-based analysis of billing activities.

QUESTION 464

What is the primary benefit of using a bridge table for multivalued dimensions?

Answer

A bridge table allows for the representation of many-to-many relationships between fact tables and dimensions, enabling flexibility in managing complex data.

QUESTION 465

What does role-enabled bottleneck authentication mean in a data warehouse context?

Answer

It refers to a security architecture where access to the data warehouse is centrally controlled by a server that authenticates users based on their roles, ensuring secure access.

QUESTION 466

What is the purpose of a real-time partition in data warehousing?

Answer

The real-time partition is designed to handle the most current data, allowing for near real-time reporting and serving as a temporary holding area for recent transactions before they are loaded into the static data warehouse.

QUESTION 467

Why is the inventory snapshot fact table denser than the sales transaction fact table?

Answer

The inventory snapshot table is denser because it contains a row for virtually every product in every store every day, unlike the sales table which only records sold products.

QUESTION 468

What are the performance advantages of using surrogate keys?

Answer

Surrogate keys are smaller, resulting in reduced storage requirements and improved query performance by minimizing the size of fact tables.

Flashcards 469 - 480

QUESTION 469

How does the design of a data warehouse facilitate historical data capture?

Answer

A data warehouse captures historical data by using dimension tables that provide context for measurements at specific points in time.

QUESTION 470

What is the impact of security and privacy on data warehousing?

Answer

Security and privacy concerns shape data warehousing practices, requiring robust measures to protect sensitive information and ensure compliance with regulations.

QUESTION 471

What is the significance of heterogeneous product schemas?

Answer

Heterogeneous product schemas allow financial services to manage diverse products with specific attributes while providing a global view of customer data across all product lines.

QUESTION 472

What is a slowly changing dimension (SCD)?

Answer

A slowly changing dimension refers to dimension attributes that change slowly over time, requiring specific strategies to track these changes in a data warehouse.

QUESTION 473

How are demographics keys used in fact tables?

Answer

Demographics keys are included as foreign keys in fact tables to efficiently access demographic information related to the facts being analyzed.

QUESTION 474

What is a Promotion in retail?

Answer

An event planned by marketing featuring causal items like ads or price reductions.

QUESTION 475

What is the impact of having late-arriving dimension rows?

Answer

Late-arriving dimension rows can complicate data updates and require careful management to ensure accurate historical representation.

QUESTION 476

How can data warehouse project managers ensure success?

Answer

Project managers can ensure success by actively managing communication, collecting business requirements, developing the technical architecture, and addressing common pitfalls throughout the project lifecycle.

QUESTION 477

What can happen if the grain is incorrectly declared?

Answer

An incorrect grain declaration can lead to difficulties in data analysis, preventing users from obtaining the insights they need.

QUESTION 478

What is the purpose of a time-of-day dimension in retail analysis?

Answer

A time-of-day dimension allows for detailed analysis of sales patterns based on different times, such as peak shopping hours or specific shifts.

QUESTION 479

What factors should be considered when developing an aggregation strategy?

Answer

Consider factors include user access patterns, statistical distribution of data, and the cost-effectiveness of building and maintaining aggregates.

QUESTION 480

How have key performance indicators (KPIs) evolved in business management?

Answer

KPIs have become more sophisticated and are derived from complex mathematical models, gaining respect in areas like economic forecasting and fraud detection, despite the risks of misinterpretation.

Flashcards 481 - 492

QUESTION 481

What are the key components of accounting in data warehousing?

Answer

Key components include general ledger data, financial statements, budgeting processes, and handling year-to-date facts across multiple fiscal calendars.

QUESTION 482

What is the purpose of a degenerate dimension?

Answer

A degenerate dimension is used when all header information associated with a transaction has been extracted into other dimensions, retaining only the transaction key in the fact table.

QUESTION 483

What is the importance of vendor performance metrics in procurement?

Answer

Vendor performance metrics, such as fill rates and on-time delivery, are crucial for assessing supplier reliability and ensuring that the procurement process supports overall business goals effectively.

QUESTION 484

What is the purpose of the date dimension in retail sales?

Answer

The date dimension provides temporal attributes necessary for analyzing sales trends over various time periods.

QUESTION 485

What is the business dimensional lifecycle?

Answer

The business dimensional lifecycle is a methodology for planning, designing, implementing, and maintaining data warehouses to ensure alignment with business needs.

QUESTION 486

What is 'procurement' in the context of supply chain management?

Answer

Procurement refers to the process of acquiring goods and services for an organization, including sourcing, purchasing, and managing supplier relationships.

QUESTION 487

What are the key performance metrics that insurance companies track?

Answer

Insurance companies track metrics such as claims processing efficiency, time lag between claim opening and first payment, and premium revenue metrics.

QUESTION 488

What are the steps involved in dimension table staging?

Answer

Steps include extracting dimensional data, cleansing attribute values, managing surrogate key assignments, and publishing revised dimensions.

QUESTION 489

What are the key metrics of a profitability data mart?

Answer

Key metrics include gross revenue, net revenue, gross profit, and net profit derived from the sales activities.

QUESTION 490

Define Gross Profit.

Answer

The gross revenue less the cost of the goods.

QUESTION 491

What is the role of metadata in a product dimension?

Answer

Metadata serves as the documentation for the definitions, interpretations, and origins of product attributes, ensuring clarity and consistency in data usage.

QUESTION 492

What is a degenerate dimension?

Answer

A degenerate dimension is a dimension key in a fact table that does not have an associated dimension table, often used for keys like POS transaction numbers.

Flashcards 493 - 504

QUESTION 493

Describe the four-step process of dimensional modeling.

Answer

The four steps are: 1) Identify the business process, 2) Identify the grain, 3) Identify the dimensions, and 4) Identify the facts.

QUESTION 494

What is a Ship Mode Dimension?

Answer

A dimension that categorizes the different methods of shipping products, helping to analyze shipping performance based on modes used.

QUESTION 495

What is the significance of primary keys in dimension tables?

Answer

Primary keys in dimension tables ensure referential integrity between the dimension and fact tables, allowing for accurate data relationships.

QUESTION 496

What is the purpose of a currency dimension in a multi-currency fact table?

Answer

A currency dimension identifies the type of currency used in transactions, enabling accurate reporting and analysis across different currency types in a multinational context.

QUESTION 497

Explain the concept of 'aggregate fact table'.

Answer

An aggregate fact table summarizes data at a higher level of granularity, making it easier to analyze trends without delving into detailed transactional data.

QUESTION 498

What is the distinction between dimension tables and fact tables?

Answer

Dimension tables contain descriptive attributes, while fact tables store quantitative data related to business processes.

QUESTION 499

What are the key processes involved in order management?

Answer

Key processes include order processing, shipment processing, and invoice processing, which generate important metrics such as sales volume and revenue.

QUESTION 500

Describe the importance of transforming data in the ETL process.

Answer

Transforming data ensures consistency, quality, and usability of the data as it moves from operational sources to the data warehouse.

QUESTION 501

What is 'demand side' in business processes?

Answer

The demand side refers to the flow of processes in a business starting with finished goods inventory and progressing to customer sales, contrasting with the supply side.

QUESTION 502

What are the challenges associated with handling multiple sources in data staging?

Answer

Challenges include reconciling data formats, ensuring data quality, and managing dependencies among different data sources.

QUESTION 503

What are some common mistakes to avoid when building a data warehouse?

Answer

Common mistakes include neglecting to understand business requirements, failing to manage scope, and not ensuring proper integration between business and IT.

QUESTION 504

What is a dimension table?

Answer

A dimension table is a structure that categorizes facts and measures in order to enable users to answer business questions, providing context to the data.

Flashcards 505 - 516

QUESTION 505

What is the significance of the accounting period in G/L?

Answer

The accounting period is essential for reconciling and finalizing financial results, allowing organizations to report accurately on their financial activities.

QUESTION 506

Define Merchandise Hierarchy.

Answer

A set of attributes in the product dimension that define an ascending many-to-one relationship, common in manufacturing and retail.

QUESTION 507

Define Business Intelligence (BI).

Answer

Business Intelligence refers to the use of an organization's internal and external information assets to make informed business decisions and improve overall performance.

QUESTION 508

Why is visitor identity challenging to establish in web analytics?

Answer

Visitors often prefer anonymity and may use different devices, making it difficult to accurately track individual identities across sessions.

QUESTION 509

Describe the concept of heterogeneous products in dimensional modeling.

Answer

Heterogeneous products represent various product types within a single fact table, allowing for a diverse set of attributes and metrics that can be analyzed collectively.

QUESTION 510

What is a City-Pair Dimension?

Answer

A dimension that describes the relationship and characteristics of flights between two airports, including distance and type of city pair.

QUESTION 511

What factors can influence the success of a data warehouse project?

Answer

Key factors include strong sponsorship, compelling business motivation, technical feasibility, and a positive relationship between business and IT.

QUESTION 512

What are common dimension tables in a retail context?

Answer

Common dimension tables in retail include product, store, customer, and time dimensions, each containing attributes relevant to the retail business.

QUESTION 513

What is the typical structure of a fact table?

Answer

A fact table typically includes foreign keys linking to dimension tables and measures that quantify the business process being analyzed.

QUESTION 514

What is a minidimension?

Answer

Minidimensions are created to store correlated attributes that are frequently updated, allowing for better tracking of attribute changes over time without overwhelming the main dimension table.

QUESTION 515

What are some examples of operational control numbers?

Answer

Examples include order numbers, invoice numbers, and bill-of-lading numbers, which often become degenerate dimensions in fact tables.

QUESTION 516

What is the role of the 'Airport Dimension' in airline data?

Answer

The Airport Dimension contains information about airports used in flights, including location and characteristics, crucial for analyzing travel patterns and performance.

Flashcards 517 - 528

QUESTION 517

Why are surrogate keys preferred over concatenated keys?

Answer

Surrogate keys avoid the complications of multiple natural keys and prevent performance issues associated with double-barreled joins.

QUESTION 518

What is the ICD-9-CM standard?

Answer

The International Classification of Diseases, 9th Revision: Clinical Modification (ICD-9-CM) is a standard used in the U.S. for coding diagnoses and procedures in health care.

QUESTION 519

What are the three fundamental types of fact tables?

Answer

The three types of fact tables are: Transaction Fact Tables, Periodic Snapshot Fact Tables, and Accumulating Snapshot Fact Tables.

QUESTION 520

What is the role of a cyclic redundancy checksum (CRC) in data staging?

Answer

CRC is used to quickly identify changes in data rows, preventing unnecessary updates for unchanged data.

QUESTION 521

What is a common issue with operational production codes?

Answer

Operational production codes can change or be recycled, leading to potential data integrity issues if used as natural keys in a data warehouse.

QUESTION 522

Why should calendar logic be stored in a dimension table rather than in application code?

Answer

Storing calendar logic in a dimension table ensures consistent values are available for reporting and avoids the complexity of decoding flags in application code, improving usability for business users.

QUESTION 523

What are some common dimensions found in data marts?

Answer

Common dimensions include date/time, transaction type, organization, employee, product, and customer.

QUESTION 524

Define 'grain' in the context of dimensional modeling.

Answer

Grain refers to the level of detail or granularity of data represented by a single row in a fact table, determining what each row describes.

QUESTION 525

What is a snapshot fact table in procurement?

Answer

A snapshot fact table captures the state of procurement data at specific points in time, allowing businesses to monitor product movement through the procurement pipeline.

QUESTION 526

What dimensions are included in the banking case study model?

Answer

Dimensions include month end date, account, household, branch, product, and status, providing a comprehensive view of the bank's operations.

QUESTION 527

What are the key dimensions typically found in a shipment invoice fact table?

Answer

Common dimensions include date, customer, product, deal, shipper, and ship-from, allowing comprehensive analysis of shipment data.

QUESTION 528

What is the importance of daily backups to removable media?

Answer

Daily backups to removable media provide essential offline security and protection against data loss, ensuring that data can be retrieved even in the event of system failures.

Flashcards 529 - 540

QUESTION 529

What is the second fact table in a clickstream data mart focused on?

Answer

The second fact table focuses on individual page events in each customer session, capturing detailed interactions such as page views and actions taken on each page.

QUESTION 530

How do health care organizations track changes in multivalued dimensions over time?

Answer

They can use date stamps within bridge tables to track when diagnosis groups change, allowing for time-sensitive queries.

QUESTION 531

How can transaction detail data be used for point-in-time balances?

Answer

Transaction detail data can help determine an account's balance at a specific historical point by tracing back to the most recent previous transaction.

QUESTION 532

Explain the term 'data granularity' in the context of fact tables.

Answer

Data granularity refers to the level of detail represented in fact tables, which influences the types of analyses that can be performed.

QUESTION 533

What is a transaction grain in a data warehouse?

Answer

Transaction grain records every individual transaction, capturing the most detailed level of data, such as each billing transaction from a provider.

QUESTION 534

What metrics can be improved by leveraging alumni data in a university context?

Answer

Metrics include targeting messages for donations, improving recruitment, job placements, and enhancing research opportunities.

QUESTION 535

Define 'customer relationship management' in the context of data warehousing.

Answer

CRM in data warehousing involves analyzing customer interactions and data throughout the customer lifecycle to improve relationships and business outcomes.

QUESTION 536

What are core product attributes?

Answer

Core product attributes are fundamental characteristics of a product that define its identity, such as name, type, and specifications.

QUESTION 537

What role do conformed dimensions play in data marts?

Answer

Conformed dimensions ensure that the same dimension is consistently defined across multiple data marts, facilitating integrated analysis.

QUESTION 538

How can the student registration event fact table be queried?

Answer

Queries can involve counting registrations by faculty, determining how many declared majors are taking specific courses, and analyzing attendance over time.

QUESTION 539

What challenges arise from having multiple source systems in procurement?

Answer

Having multiple source systems complicates data integration and staging processes, making it challenging to build a cohesive view of procurement transactions, which may necessitate separate fact tables.

QUESTION 540

How does the increasing volume of data affect data warehousing?

Answer

The increasing volume of data necessitates more robust storage solutions, advanced processing capabilities, and efficient data management practices to derive actionable insights.

Flashcards 541 - 552

QUESTION 541

What does the term 'grain' refer to in a fact table?

Answer

The 'grain' of a fact table refers to the level of detail at which measurements are recorded, determining the combination of dimensions that define each measurement.

QUESTION 542

Define 'data mart' in the context of data warehousing.

Answer

A data mart is a subset of a data warehouse, often focused on a specific business line or team, designed to meet particular analytical needs.

QUESTION 543

What is the importance of the warehouse team in managing surrogate keys?

Answer

The warehouse team maintains control over the data environment, ensuring that operational changes do not disrupt data integrity and access.

QUESTION 544

What is the significance of using a centralized authority for dimension management?

Answer

A centralized authority ensures consistency and conformance of dimensions across different data marts, reducing redundancy.

QUESTION 545

What is the difference between a forward proxy and a reverse proxy?

Answer

A forward proxy is outside control and serves requests from users, while a reverse proxy is controlled by the server and helps offload requests.

QUESTION 546

What is the benefit of using multiple hierarchies in a dimension table?

Answer

Using multiple hierarchies allows for a more versatile analysis, enabling users to explore data across different contexts, such as geographic and product categories simultaneously.

QUESTION 547

What does the term 'type 2 slowly changing dimension' refer to?

Answer

Type 2 slowly changing dimension involves tracking historical changes in data by creating new records for each change while retaining the original record, allowing for accurate historical analysis.

QUESTION 548

How does the design of a health care billing line item fact table support profitability analysis?

Answer

By including additional facts such as consumables cost, provider cost, and net profit, the billing line item fact table can provide a comprehensive view of profitability.

QUESTION 549

How does dimensional modeling address the need for both global and line-of-business views?

Answer

Dimensional modeling provides a global view through core fact tables while offering specific line-of-business views with custom schemas tailored to unique attributes.

QUESTION 550

What is the role of analysts in financial reporting?

Answer

Analysts interpret financial data, produce reports, and provide insights that guide management decisions and strategic planning.

QUESTION 551

Describe the Type 1 approach for handling slowly changing dimensions.

Answer

The Type 1 approach overwrites the previous dimension attribute value with the most current descriptor, assuming all changes are corrections.

QUESTION 552

What role do promotions play in retail sales?

Answer

Promotions, such as price reductions and advertisements, are crucial for driving sales and can significantly impact customer purchasing behavior.

Flashcards 553 - 564

QUESTION 553

Why are invoice fact tables considered powerful?

Answer

They combine essential business elements such as customers, products, and profitability factors, offering a comprehensive view of business operations.

QUESTION 554

What role does user feedback play in the development of a data warehouse?

Answer

User feedback is essential for refining requirements and ensuring that the warehouse meets the evolving needs of the organization, guiding development and prioritization.

QUESTION 555

What is the role of metadata in the technical architecture of a data warehouse?

Answer

Metadata serves as a crucial component for data management, providing context, definitions, and guidance for data users and systems within the warehouse.

QUESTION 556

What are some common challenges in data staging?

Answer

Common challenges include data quality issues, handling missing or inconsistent values, and managing complex transformations.

QUESTION 557

What is the purpose of a bus matrix in data warehousing?

Answer

A bus matrix outlines the business processes and the dimensions associated with each process, helping to visualize and plan the architecture of a data warehouse.

QUESTION 558

What does MBA stand for?

Answer

Master of Business Administration, a graduate degree focused on understanding commercial business organization and management.

QUESTION 559

What is the significance of having a registration count in a factless fact table?

Answer

Including a registration count allows for easier SQL queries and clearer data representation, even if it does not add substantive information.

QUESTION 560

How can organizations ensure that their data warehouse projects remain relevant?

Answer

Organizations should continuously assess and adjust their data warehouse projects based on changing business needs, technological advances, and user feedback.

QUESTION 561

What is a SAN (Storage Area Network)?

Answer

A SAN is a cluster of high-performance disk drives and backup devices connected via high-speed fiber channel technology, allowing for efficient disk-to-disk backup and access by multiple application systems in parallel.

QUESTION 562

What is the impact of dimensional model grain on adding new dimensions?

Answer

The grain defines the level of detail; a finer grain allows for easier addition of new dimensions without disrupting existing schema.

QUESTION 563

What is the role of multiple units of measure in order management?

Answer

Different functional areas may require performance metrics in various units, necessitating clear conversion factors to ensure accurate reporting across departments.

QUESTION 564

Describe the typical attributes of a product dimension table.

Answer

A product dimension table typically includes attributes like product key, product description, SKU number, brand description, category description, and packaging type.

Flashcards 565 - 576

QUESTION 565

What are the source systems involved in procurement transactions?

Answer

The source systems involved are a purchasing system for purchase requisitions and purchase orders, a warehousing system for shipping notices and warehouse receipts, and an accounts payable system for vendor payments.

QUESTION 566

What is the significance of the 'rate plan dimension' in telecommunications?

Answer

The rate plan dimension describes the pricing structures for services, detailing different plans available to customers, which aids in revenue analysis.

QUESTION 567

What role do surrogate keys play in dimensional modeling?

Answer

Surrogate keys serve as unique identifiers in fact and dimension tables, enabling efficient joins and maintaining data integrity across tables.

QUESTION 568

What are some key performance metrics that can be derived from CRM data?

Answer

Key performance metrics include sales effectiveness, customer retention rates, customer satisfaction scores, and profitability margins.

QUESTION 569

What types of analyses are necessary for health care billing data warehouses?

Answer

Necessary analyses include tracking counts and amounts of bills, payment statuses, and time taken for payments.

QUESTION 570

What constitutes the customer ship-to dimension?

Answer

The customer ship-to dimension includes distinct locations for product shipments, with attributes such as Customer Ship To Name, Address, City, State, and Zip Code.

QUESTION 571

What challenges may arise from using non-integrated data marts?

Answer

Non-integrated data marts can lead to inconsistencies in reporting and analysis, creating a fragmented view of organizational data and hindering decision-making.

QUESTION 572

Define the concept of heterogeneous products in insurance.

Answer

Heterogeneous products refer to the various lines of insurance that have different attributes and coverage types, necessitating custom dimension tables for specific coverages.

QUESTION 573

What challenges can arise during CRM implementation?

Answer

Challenges include resistance to change, ineffective existing processes, the need for organizational commitment, and the necessity for a clear business strategy to guide the implementation.

QUESTION 574

Why is the periodic snapshot grain not suitable for health care billing?

Answer

The periodic snapshot does not effectively capture the rapid changes and behaviors surrounding medical bills, as most activity occurs quickly.

QUESTION 575

What is the primary goal of Customer Relationship Management (CRM)?

Answer

The primary goal of CRM is to maximize relationships with customers over their lifetime by understanding their needs and maintaining valuable, long-lasting connections.

QUESTION 576

What is the role of human resources management in a data warehouse?

Answer

Human resources management in a data warehouse involves tracking employee data, managing recruitment processes, and analyzing workforce metrics for decision support.

Flashcards 577 - 588

QUESTION 577

Explain the concept of aggregation in data warehousing.

Answer

Aggregation involves creating precalculated and prestored tables to enhance query performance by summarizing detailed data.

QUESTION 578

What is the importance of the product dimension?

Answer

The product dimension is crucial as it provides detailed information about the products sold, including numerous descriptors and hierarchies for analytical purposes.

QUESTION 579

What are multivalued dimensions in a healthcare data model?

Answer

Multivalued dimensions are those that can hold multiple values for a single record, such as a patient having multiple diagnoses.

QUESTION 580

Why is it important to maintain a cross-reference table for surrogate keys?

Answer

A cross-reference table is essential for substituting appropriate surrogate keys in fact and dimension tables, ensuring data integrity and consistency.

QUESTION 581

What is the primary key of the retail sales fact table?

Answer

The primary key consists of the degenerate POS transaction number and the product key, which ensures uniqueness for each transaction line item.

QUESTION 582

What is the role of a status dimension in a financial services model?

Answer

The status dimension captures the current condition of an account, such as whether it is active or inactive, and any changes that occurred during the month.

QUESTION 583

What role does the data presentation area play?

Answer

The data presentation area organizes and stores data for direct querying by users, report writers, and analytical applications, effectively acting as the interface for business users.

QUESTION 584

What are the key challenges in inventory management?

Answer

Key challenges in inventory management include balancing supply and demand, minimizing costs, managing stock levels, and ensuring product availability.

QUESTION 585

What is the purpose of the G/L account dimension?

Answer

The G/L account dimension provides detailed attributes for each G/L account, enabling more granular financial analysis and reporting.

QUESTION 586

Give examples of grain declarations.

Answer

Examples include an individual line item on a retail sales ticket, a line item on a medical bill, an individual boarding pass, and daily inventory snapshots.

QUESTION 587

What political forces are currently affecting data warehousing?

Answer

Security and privacy issues are significant political forces impacting data warehousing, requiring solutions that address user concerns while managing data responsibly.

QUESTION 588

What are common terms found in a glossary related to dimensional modeling?

Answer

Common terms include Fact Table, Dimension Table, Schema, Star Schema, Snowflake Schema, and Role-Playing Dimension, which are essential for understanding data warehousing concepts.

Flashcards 589 - 600

QUESTION 589

What is the significance of dimensional modeling in data warehousing?

Answer

Dimensional modeling structures data in a way that simplifies analysis and reporting, making it easier for business users to understand and use data.

QUESTION 590

What are fixed-depth hierarchies in dimensional modeling?

Answer

Fixed-depth hierarchies are organizational structures with a predictable number of levels, such as a maximum of three roll-up levels like corporate parent, business unit headquarters, and regional offices. Each level corresponds to distinct attributes in the customer dimension.

QUESTION 591

What is the purpose of the referral dimension?

Answer

The referral dimension describes how a customer arrived at a page, including the referring URL, site, and search type.

QUESTION 592

What is 'Transactional Fact Table'?

Answer

A Transactional Fact Table is a type of fact table where each row corresponds to a single transaction event, capturing detailed transactional data.

QUESTION 593

What is the role of financial analysts in an organization?

Answer

Financial analysts are responsible for analyzing financial data, disseminating insights, and ensuring that managers have timely access to key financial metrics.

QUESTION 594

Discuss the importance of date milestones in the applicant tracking process.

Answer

Date milestones allow tracking of the applicant's progress through the admissions process, helping identify bottlenecks and analyze pacing.

QUESTION 595

Define the term 'surrogate key' in dimensional modeling.

Answer

A surrogate key is a sequentially assigned integer used to join dimension tables to fact tables, replacing natural keys to avoid operational dependencies.

QUESTION 596

What is the significance of the Extract-Transform-Load (ETL) process?

Answer

The ETL process is crucial for transferring data from operational systems to the data warehouse, involving extraction of data, transformation for consistency and quality, and loading into the presentation area.

QUESTION 597

Describe the grain of a fact table in the context of student registration events.

Answer

The grain of the fact table for student registration events is defined as one row for each registered course by student and term, allowing for detailed tracking of registrations.

QUESTION 598

What are junk dimensions?

Answer

Junk dimensions are used to group together various correlated indicators and flags that do not fit neatly into other dimensions, simplifying the model and enhancing clarity.

QUESTION 599

Define the term 'grain' in the context of a fact table.

Answer

Grain refers to the level of detail represented in a fact table, determining the unique combination of dimensions that make up each record.

QUESTION 600

What is the significance of variance analysis in financial management?

Answer

Variance analysis helps organizations understand the differences between budgeted and actual financial outcomes, enabling corrective actions.

Flashcards 601 - 612

QUESTION 601

What is a bridge table in a data warehouse?

Answer

A bridge table captures many-to-many relationships between fact and dimension tables, enabling complex hierarchies or dimensions that cannot be represented by a single table.

QUESTION 602

What attributes are typically included in a date dimension?

Answer

Attributes may include date key, full date description, day of the week, month number, fiscal year, holiday indicator, and season.

QUESTION 603

What is the significance of claim reopening and reserve resetting?

Answer

These measures indicate the frequency and circumstances under which claims are reopened and reserves are adjusted, which is an important metric for insurance companies.

QUESTION 604

Explain variable-depth hierarchies.

Answer

Variable-depth hierarchies represent unpredictable organizational structures where the number of levels can vary. This complexity requires advanced techniques to manage and summarize data across these levels.

QUESTION 605

What is the importance of the date dimension in retail sales?

Answer

The date dimension is critical as it allows analysis of sales over time, enabling businesses to track trends, seasonal variations, and performance metrics.

QUESTION 606

How can organizations ensure that their customer dimensions do not grow too large?

Answer

Organizations can manage the size of customer dimensions by breaking off frequently changing attributes into minidimensions, maintaining performance and usability.

QUESTION 607

What are the limitations of using surveys for gathering requirements?

Answer

Surveys are often too simplistic, do not allow for deep probing, and fail to build rapport between users and the data warehousing initiative.

QUESTION 608

What is a surgical events transaction fact table?

Answer

A surgical events transaction fact table records data related to surgical procedures, allowing analysis of profit and loss (P&L) associated with each surgery.

QUESTION 609

What is the significance of maintaining a limited data retention policy in a data mart?

Answer

A limited data retention policy helps in managing storage efficiently by keeping only relevant data, which is crucial for performance and cost-effectiveness in data warehousing.

QUESTION 610

What dimensions are included in the student registration event fact table?

Answer

The dimensions include term, student, declared major, credit attainment, course, and faculty.

QUESTION 611

Define Referential Integrity (RI).

Answer

A condition where all keys in the fact tables are legitimate foreign keys relative to the dimension tables.

QUESTION 612

Explain the significance of the POS transaction number in retail sales.

Answer

The POS transaction number acts as a grouping key for linking products purchased in a transaction, even though it lacks descriptive attributes.

Flashcards 613 - 624

QUESTION 613

Explain the four-step dimensional design process.

Answer

The four-step dimensional design process includes: 1. Select the Business Process, 2. Declare the Grain, 3. Choose the Dimensions, and 4. Identify the Facts.

QUESTION 614

What are financial statements, and what role does the G/L play in their creation?

Answer

Financial statements like balance sheets and income statements are derived from the G/L, which provides the necessary financial data for accurate reporting.

QUESTION 615

What does the audit dimension in a fact table include?

Answer

The audit dimension includes metadata such as source system information, transformation status, and confidence indicators for data quality.

QUESTION 616

What is the significance of conformed dimensions in data warehousing?

Answer

Conformed dimensions ensure consistency across the data warehouse by allowing the same dimension to be used across multiple fact tables, facilitating better data integration and reporting.

QUESTION 617

What type of information is typically captured in HR transactional data?

Answer

HR transactional data captures employee demographics, skills, earnings, performance, and changes to employee profiles.

QUESTION 618

What is a fact dimension?

Answer

A fact dimension is a special type of dimension used to handle sparse or heterogeneous measurements in a fact table, allowing for flexibility and efficient querying.

QUESTION 619

What is the significance of the 'referrer' in web analytics?

Answer

The referrer indicates the source of a visitor's traffic, which is critical for analyzing marketing effectiveness and billing for advertising impressions.

QUESTION 620

How are changes in attributes tracked in financial services dimensional modeling?

Answer

Changes are tracked using minidimensions and bridge tables, rather than relying on type 2 slowly changing dimensions, allowing for efficient data management.

QUESTION 621

What is data warehousing?

Answer

Data warehousing refers to the process of collecting and managing data from various sources to provide meaningful business insights. It is central to IT's responsibility to make useful data available.

QUESTION 622

What are the implications of using distinct control numbers in procurement transactions?

Answer

Distinct control numbers for procurement transactions indicate separate processes, suggesting that multiple fact tables may be necessary to accurately represent and analyze each distinct transaction type.

QUESTION 623

What two primary techniques are used for gathering business requirements?

Answer

The two primary techniques are interviews and facilitated sessions, each with its own advantages and disadvantages.

QUESTION 624

What is the general ledger (G/L)?

Answer

The general ledger is a core financial system that integrates detailed information from subledgers such as purchasing, payables, and receivables.

Flashcards 625 - 636

QUESTION 625

How can SQL be used with a factless fact table?

Answer

SQL can be used to count keys or unique instances within the factless fact table to analyze relationships between dimensions without needing numerical facts.

QUESTION 626

What are the goals of a data warehouse?

Answer

The goals include making data easily accessible, presenting information consistently, being adaptive to change, ensuring security, and serving as a foundation for improved decision-making.

QUESTION 627

What is the primary benefit of using surrogate keys?

Answer

Surrogate keys buffer the data warehouse environment from operational changes, allowing the warehouse team to maintain control over the data.

QUESTION 628

What are the key considerations for the dimensionality of facts in procurement?

Answer

Considerations include whether the same dimensions apply across transaction types and whether unique attributes exist for different processes.

QUESTION 629

How does the addition of new measured facts affect a fact table?

Answer

New measured facts can be added by modifying the fact table to include new columns or creating a new fact table if the grain differs.

QUESTION 630

What is the purpose of a data warehouse support team?

Answer

The support team is responsible for ongoing maintenance and user assistance, ensuring the data warehouse remains functional and responsive to user needs.

QUESTION 631

What is the importance of a strong business sponsor?

Answer

A strong business sponsor provides vision, advocacy, and political support, essential for navigating organizational challenges and ensuring project alignment.

QUESTION 632

Define a data mart in the context of a data warehouse.

Answer

A data mart is a subset of the data warehouse that presents data from a single business process, providing a focused view for specific business needs.

QUESTION 633

How does improved access to financial data benefit organizations?

Answer

Improved access to financial data enhances risk management, streamlines operations, and identifies cost savings, enabling better decision-making across the organization.

QUESTION 634

What does HCPCS stand for and what is its purpose?

Answer

The Health Care Financing Administration Common Procedure Coding System (HCPCS) is a set of health care procedure codes used for billing health-related services and items.

QUESTION 635

How can organizations combine customer data from multiple sources?

Answer

Organizations can combine customer data by integrating various data sources into a unified system, ensuring consistency and accuracy for analysis and reporting.

QUESTION 636

What are some examples of roles played by the date dimension in health care billing?

Answer

Examples include treatment date, primary insurance billing date, and last payment date, each serving a distinct purpose in analyzing billing history.

Flashcards 637 - 648

QUESTION 637

What is the role of a date dimension in order management?

Answer

The date dimension is essential for analyzing order management performance over time; it provides context for transactions by linking to multiple date attributes.

QUESTION 638

What is the difference between multiple-transaction and single-transaction fact tables?

Answer

Multiple-transaction fact tables store data from various transactions related to a business process, while single-transaction fact tables focus on a single event or transaction type.

QUESTION 639

Explain the concept of a 'dirty customer dimension'.

Answer

A dirty customer dimension is a customer dimension where the same individual may appear multiple times with variations in name spellings or attributes, complicating customer identification.

QUESTION 640

What is a Join Constraint in SQL?

Answer

The portion of the SQL WHERE clause that maintains the join relationships between fact and dimension tables.

QUESTION 641

What is the significance of salvage payments in claims?

Answer

Salvage payments represent the value recovered from a covered item after the insurance company replaces it, and they are credited against the claim accounting.

QUESTION 642

What is an invoice transaction in a manufacturing context?

Answer

An invoice transaction occurs when products are shipped, detailing the products and associated prices, discounts, and allowances for each line item.

QUESTION 643

What characteristics define good dimension attributes?

Answer

Good dimension attributes are textual, discrete, meaningfully descriptive, and allow for robust query constraints and reporting.

QUESTION 644

How can claims data be analyzed for performance measurement?

Answer

Claims data can be analyzed by rolling forward all transactions to update an accumulating claims snapshot, which provides a clear view of claim performance without detailed transaction analysis.

QUESTION 645

What is the significance of the budget line item?

Answer

A budget line item specifies the purpose and amount allocated for spending within a budget, allowing organizations to track financial commitments.

QUESTION 646

What is a critical consideration when separating fact data by granularity?

Answer

Fact data must be separated early in the staging process to ensure that data at different levels of detail is handled appropriately.

QUESTION 647

What is the difference between Operational Data Store (ODS) and data warehouse in CRM?

Answer

The ODS is designed for real-time operational support and contains a subset of data for immediate use, while the data warehouse is a comprehensive repository for historical customer data.

QUESTION 648

What is the main takeaway from the chapter on Present Imperatives and Future Outlook?

Answer

The chapter highlights future trends in data warehousing, emphasizing the need for adaptability to changing user requirements and technological advancements.

Flashcards 649 - 660

QUESTION 649

How can organizations ensure compliance with privacy regulations?

Answer

Organizations can ensure compliance by regularly auditing data practices, implementing clear privacy policies, and training staff on legal obligations regarding data handling.

QUESTION 650

What is a Stock Keeping Unit (SKU)?

Answer

A standard term in manufacturing and retail to describe an individual product, often used for inventory management.

QUESTION 651

How can we measure the time spent by a customer during a session?

Answer

The time spent is measured in 'session seconds', which is the total duration a customer spends on the site during that session.

QUESTION 652

Why is it important to standardize labels in dimension tables?

Answer

Standardizing labels ensures consistent reporting across applications and facilitates easier data analysis and interpretation.

QUESTION 653

What is the purpose of including both universal and local date/time dimensions?

Answer

Universal date/time dimensions ensure synchronization across multiple time zones, while local date/time dimensions reflect the actual wall clock time for the customer, aiding in accurate session tracking.

QUESTION 654

Why should natural keys be avoided in data warehouses?

Answer

Natural keys can lead to dependencies on operational codes which may change, making queries and data access vulnerable to invalidation.

QUESTION 655

What is the purpose of conformed dimensions in a data warehouse?

Answer

Conformed dimensions allow different fact tables to share the same dimensions, ensuring consistent data analysis across various data sources.

QUESTION 656

What is an Operational Data Store (ODS)?

Answer

A set of tables providing immediate reporting of operational results, sitting between operational systems and the data warehouse.

QUESTION 657

What is the significance of a transaction-level fact table?

Answer

A transaction-level fact table captures detailed transaction data, allowing for granular analysis of sales and consumer behavior.

QUESTION 658

How can new attributes be added to existing dimensions?

Answer

New attributes can be added as new columns in the dimension table without affecting existing applications.

QUESTION 659

What is the definition of business intelligence (BI)?

Answer

Business intelligence (BI) refers to leveraging an organization’s internal and external information assets to make better business decisions.

QUESTION 660

What types of transactions are captured by the operational policy transaction system?

Answer

The system captures transactions such as creating, altering, and canceling policies; creating, altering, and canceling coverage; and underwriting policies.

Flashcards 661 - 672

QUESTION 661

What is the significance of the ship-from dimension?

Answer

The ship-from dimension provides details about the manufacturing warehouse or shipping location, which is important for logistics analysis.

QUESTION 662

What types of businesses are included in the financial services industry?

Answer

The financial services industry includes a variety of businesses such as credit card companies, brokerage firms, mortgage providers, and retail banks.

QUESTION 663

How should name and address attributes be designed in CRM?

Answer

Name and address attributes should be broken down into elemental parts for better analysis and data quality, rather than using general-purpose fields.

QUESTION 664

What is the significance of OLAP cubes in data warehousing?

Answer

OLAP cubes enhance performance for summarized data access and facilitate rapid response to analytical queries.

QUESTION 665

What is the importance of tracking changes in large customer dimensions?

Answer

Tracking changes in large customer dimensions is crucial for maintaining accurate customer profiles and ensuring effective decision-making, especially in industries like insurance.

QUESTION 666

How should data warehouses prepare for evolving technologies?

Answer

Data warehouses should adopt flexible architectures that can quickly adapt to new technologies and changing user demands, ensuring they remain relevant and functional.

QUESTION 667

What is the importance of audit dimensions?

Answer

Audit dimensions track the lineage of data, providing transparency and accountability in data management.

QUESTION 668

What is a Fact Table?

Answer

A table in a data warehouse that stores quantitative data for analysis and is typically linked to dimension tables.

QUESTION 669

What is a web client-server interaction?

Answer

A web client-server interaction occurs when a browser requests information from a web server, typically using the HTTP protocol.

QUESTION 670

What is the purpose of the policy premium snapshot fact table?

Answer

The policy premium snapshot fact table captures the cumulative effect of premium transactions over time, providing a summary of premium revenue metrics.

QUESTION 671

What is a custom schema in financial services?

Answer

A custom schema is designed for a specific line of business to capture unique facts and attributes, avoiding the sparsity of a core schema filled with nulls.

QUESTION 672

Explain the concept of allocating header facts to line items.

Answer

Allocating header facts to line items allows for a more detailed analysis by distributing costs and other metrics associated with an entire order across individual line items.

Flashcards 673 - 684

QUESTION 673

What does it mean to treat an attribute as a Type 1 change?

Answer

Treating an attribute as a Type 1 change means overwriting the existing value with the new value without creating a new row, which simplifies the data model but loses historical data.

QUESTION 674

What is the primary focus of a data warehouse?

Answer

The primary focus of a data warehouse is to meet the needs of business users by presenting data in a simple, understandable dimensional format, facilitating decision-making processes.

QUESTION 675

What is the significance of banding in minidimensions?

Answer

Banding reduces the number of combinations in a minidimension by forcing continuously variable attributes, such as income, into discrete ranges, thus simplifying the data structure.

QUESTION 676

What does ROI stand for?

Answer

Return on Investment, typically expressed as a rate describing the growth of an investment over time.

QUESTION 677

Why is access to financial data important for managers?

Answer

Timely access to financial data allows managers to analyze performance trends, variances, and anomalies quickly, fostering fiscal responsibility and accountability.

QUESTION 678

What are some common pitfalls to avoid in dimensional modeling?

Answer

Common pitfalls include overly complex designs, failing to involve business users in the design process, and neglecting to address slowly changing dimensions effectively.

QUESTION 679

What are some reasons for including a 'No Promotion in Effect' row in the promotion dimension?

Answer

Including a 'No Promotion in Effect' row avoids referential integrity issues in the fact table and prevents confusion in user queries that could arise from null promotion keys.

QUESTION 680

How does CRM promise significant returns for organizations?

Answer

CRM can lead to increased sales effectiveness, revenue growth, enhanced sales productivity, improved customer profitability, higher customer satisfaction, and increased customer retention.

QUESTION 681

Why is clickstream data considered complex?

Answer

Clickstream data is complex due to various log formats, the need for synchronization across servers, and the presence of ambiguities regarding visitor identity and session tracking.

QUESTION 682

What is the role of the policyholder dimension?

Answer

The policyholder dimension captures customer information which can include individuals or businesses purchasing insurance policies.

QUESTION 683

In the retail case study, what business process is modeled?

Answer

The business process modeled is POS retail sales, allowing analysis of what products sell in which stores under various conditions.

QUESTION 684

Define the term 'fact table'.

Answer

A fact table is a central table in a dimensional model that contains quantitative data for analysis and is typically linked to dimension tables.

Flashcards 685 - 696

QUESTION 685

What is the purpose of tracking changes in customer dimensions?

Answer

Tracking changes in customer dimensions is crucial for businesses like insurance companies to maintain accurate and up-to-date information about their customers for policy approvals and claims.

QUESTION 686

What is the Referral Dimension?

Answer

The Referral Dimension describes how a customer arrived at the current page, including the referring URL and type (e.g., search engine, intra-site). It helps understand the source of traffic and customer behavior.

QUESTION 687

Define the term 'fact table' in the context of data warehousing.

Answer

A fact table is a central table in a star schema of a data warehouse that contains quantitative data for analysis and is typically associated with dimension tables detailing context.

QUESTION 688

What is the single most important design technique in dimensional modeling?

Answer

The most important design technique is conforming dimensions across fact tables to enable seamless data integration and accurate analysis.

QUESTION 689

How does electronic commerce affect data warehousing?

Answer

Electronic commerce generates large volumes of transactional data that must be effectively captured, stored, and analyzed within a data warehouse to support business insights and decision-making.

QUESTION 690

What does declaring the grain of a business process mean?

Answer

Declaring the grain means specifying what an individual fact table row represents, conveying the level of detail associated with the fact measurements.

QUESTION 691

What is the significance of Time Stamping in databases?

Answer

Time Stamping involves tagging each record with the time when the data was processed or stored, which is crucial for tracking changes and managing data history.

QUESTION 692

Explain the concept of atomic data in dimensional modeling.

Answer

Atomic data refers to the most detailed data collected in a business process, allowing for maximum analytical flexibility and detailed analysis.

QUESTION 693

What is a fact table in a data warehouse?

Answer

A fact table stores quantitative data for analysis, typically containing foreign keys to dimension tables and metrics like sales amounts and quantities sold.

QUESTION 694

What is the purpose of documenting product attribute definitions in metadata?

Answer

Documenting attribute definitions in metadata provides a comprehensive reference for understanding the data warehouse's content, akin to an encyclopedia for users.

QUESTION 695

What is the difference between Transaction Fact Table and Periodic Snapshot Fact Table?

Answer

A Transaction Fact Table has one row for each transaction event, while a Periodic Snapshot Fact Table represents performance at regular time periods.

QUESTION 696

What is the Session Dimension in a clickstream data mart?

Answer

The Session Dimension provides levels of diagnosis for a visitor's session, capturing both local context (e.g., requesting product information) and overall context (e.g., ordering a product). It helps analyze customer behavior by labeling customer statuses such as 'high-value reliable customer' or 'new customer'.

Flashcards 697 - 708

QUESTION 697

What does the data staging area do?

Answer

The data staging area serves as a temporary storage space where data is cleansed, transformed, and prepared before being loaded into the data warehouse's presentation area.

QUESTION 698

What is the function of an analytic application in data warehousing?

Answer

Analytic applications are prebuilt data access tools intended for less frequent users, allowing for flexible analysis of data through parameter-driven queries.

QUESTION 699

What is the difference between a treatment billing record and a treatment medical record?

Answer

A treatment billing record corresponds to line items on bills sent to payers, while a treatment medical record contains comprehensive details about patient treatments, including tests and notes.

QUESTION 700

What should be included in the executive summary of the findings document?

Answer

The executive summary should provide an overview of the project, the process used for gathering requirements, and a summary of key findings and recommendations.

QUESTION 701

Why should a designer be cautious about mixing different granularities in a single fact table?

Answer

Mixing different granularities can lead to confusion and inaccuracies in data analysis. It's essential to separate different levels of facts to ensure clarity and integrity.

QUESTION 702

What is a factless fact table and why is it used in retail sales?

Answer

A factless fact table records relationships between keys without measurement metrics, such as a promotion coverage table that tracks products on promotion regardless of sales.

QUESTION 703

What is a common mistake to avoid in data warehouse development?

Answer

Avoid assuming that business users will naturally gravitate toward robust data and develop their own analytic applications; they need prebuilt applications to engage with the data effectively.

QUESTION 704

What is the significance of customer satisfaction metrics in order management?

Answer

Customer satisfaction metrics track performance such as on-time shipping and damage-free delivery, which are crucial for evaluating order fulfillment effectiveness.

QUESTION 705

Define Heterogeneous Products.

Answer

A set of products characterized by many incompatible attributes and measurable facts, posing design challenges in financial services.

QUESTION 706

What is a slowly changing dimension (SCD)?

Answer

A slowly changing dimension is a dimension that changes slowly over time, and it is important for tracking historical changes in attributes.

QUESTION 707

Why is it important to declare the grain of a business process?

Answer

Declaring the grain is crucial as it defines the level of detail for the fact table, ensuring all stakeholders have a shared understanding of the data's representation.

QUESTION 708

What is the role of the architecture task force?

Answer

The architecture task force, typically consisting of key technical personnel, focuses on designing the technical architecture, establishing its importance, and guiding the project team in its implementation.

Flashcards 709 - 720

QUESTION 709

How should late-arriving fact rows be handled in a data warehouse?

Answer

Late-arriving fact rows should be inserted into their correct historical position using date stamps from dimension rows to maintain accurate historical records.

QUESTION 710

What is the Type 1 response to slowly changing dimensions?

Answer

The Type 1 response involves overwriting the old attribute value in the dimension row with the current value, losing all historical data.

QUESTION 711

What is a URL?

Answer

Uniform Resource Locator, the text address of a specific object on the World Wide Web.

QUESTION 712

What is the importance of 'order management' in business operations?

Answer

Order management is critical for tracking customer orders from placement through fulfillment, ensuring accuracy and efficiency in delivery and customer satisfaction.

QUESTION 713

What is the significance of using cookies in tracking sessions?

Answer

Cookies help maintain a visitor's session across multiple page requests, allowing for more accurate tracking of visitor behaviors.

QUESTION 714

What is the importance of the technical architecture in a data warehouse?

Answer

Technical architecture provides the framework for integrating multiple technologies and supports the overall functionality of the data warehouse.

QUESTION 715

What are some of the metrics recorded in the monthly account snapshot fact table?

Answer

Metrics include primary balance, average daily balance, number of transactions, interest paid, interest charged, and fees charged.

QUESTION 716

What is a Type 2 slowly changing dimension?

Answer

A Type 2 slowly changing dimension creates a new row in the dimension table to capture historical changes, allowing for the tracking of changes over time without losing the old data.

QUESTION 717

What mistakes should be avoided in dimensional modeling?

Answer

Common mistakes include placing text attributes in fact tables, limiting descriptive attributes in dimensions, splitting hierarchies across dimensions, ignoring the need to track dimension attribute changes, and failing to conform facts and dimensions across separate fact tables.

QUESTION 718

What challenges might arise during a data warehouse deployment?

Answer

Challenges include data quality issues, user resistance, insufficient training, and failure to meet business needs, which can undermine the success of the deployment.

QUESTION 719

What role does electronic commerce play in modern business?

Answer

Electronic commerce facilitates online buying and selling of goods and services, leveraging digital platforms to enhance customer reach and operational efficiency.

QUESTION 720

What are the benefits of having a separate date dimension?

Answer

A separate date dimension allows for easy filtering and analysis by various date attributes, such as weekdays, holidays, and fiscal periods.

Flashcards 721 - 732

QUESTION 721

Explain the concept of householding in banking.

Answer

Householding involves grouping multiple accounts that belong to individuals within the same economic unit, allowing banks to analyze relationships and market products effectively.

QUESTION 722

What does 'up-selling' mean?

Answer

Up-selling is the practice of encouraging customers to purchase more expensive items or upgrades compared to their initial selection, enhancing sales per transaction.

QUESTION 723

What is the importance of the transaction expiration date in HR data?

Answer

The transaction expiration date helps define the time span during which the employee profile is accurate, facilitating precise queries.

QUESTION 724

What are the implications of having multiple fiscal accounting calendars?

Answer

Multiple fiscal accounting calendars can complicate financial reporting and analysis, requiring careful alignment of data and consistent application of accounting principles across different periods.

QUESTION 725

What is an accumulating snapshot fact table?

Answer

An accumulating snapshot fact table captures the state of a business process over time, reflecting the ongoing status of events rather than just historical data.

QUESTION 726

What is the purpose of banding attribute values in minidimensions?

Answer

Banded values reduce the complexity of tracking discrete values by grouping them into ranges, such as income or profitability scores, which supports routine analytic requirements.

QUESTION 727

What are some key performance indicators (KPIs) in financial analysis?

Answer

KPIs may include metrics like revenue growth, profit margins, and return on investment, helping organizations assess financial health.

QUESTION 728

How does a customer relationship management (CRM) system function?

Answer

CRM systems focus on operational and analytic processes to better understand and service customers, maximizing mutually beneficial relationships.

QUESTION 729

What is the purpose of the four-step dimensional design process?

Answer

The four-step process guides the design of dimensional models, ensuring that the resulting database structure accurately reflects business processes and user needs.

QUESTION 730

What is a dimension table?

Answer

A dimension table contains descriptive attributes related to the facts, allowing for rich context and categorization of the quantitative data stored in the fact table.

QUESTION 731

What are the phases of architecture implementation?

Answer

Architecture implementation phases include identifying mandatory capabilities, documenting the architecture plan, and determining which aspects can be deferred.

QUESTION 732

What are some examples of electronic commerce in the context of data warehousing?

Answer

Examples include online shopping platforms, personalized marketing through purchase behavior analysis, and e-learning services delivered via data warehouses.

Flashcards 733 - 744

QUESTION 733

What does 'demographic minidimension' refer to?

Answer

A demographic minidimension is a set of attributes that provides detailed descriptive information about a demographic aspect within a dimensional model.

QUESTION 734

Why should organizations focus on first-level data marts before consolidated data marts?

Answer

Focusing on first-level data marts as dimensional building blocks simplifies the ETL process and builds foundational success before tackling the complexities of consolidating multiple sources into a single model.

QUESTION 735

What are the three basic techniques for handling slowly changing dimensions?

Answer

The three basic techniques are Type 1 (overwrite the value), Type 2 (add a dimension row), and Type 3 (add a dimension column).

QUESTION 736

Define 'Web' in the context of internet technology.

Answer

The Web, or World Wide Web, is a collection of servers and browsers that communicate via the HTTP protocol, enabling access to web pages and resources.

QUESTION 737

What is the significance of a surrogate key in a product dimension?

Answer

A surrogate key is a unique identifier used in the product dimension to avoid issues with duplicate keys and to facilitate changes in product attributes over time.

QUESTION 738

What is the significance of biometric identification in data warehousing?

Answer

Biometric identification provides a secure method for verifying user identity, essential for ensuring privacy and security of personal data stored in data warehouses.

QUESTION 739

What is the function of a cost center in financial management?

Answer

A cost center is a unit within an organization that tracks expenses; it helps in budgeting and financial accountability.

QUESTION 740

What is the significance of user education in data warehouse deployment?

Answer

User education helps ensure acceptance of the warehouse by the business community and improves their ability to effectively utilize the data and applications provided.

QUESTION 741

What is the significance of the customer dimension's foreign keys?

Answer

Foreign keys in the customer dimension link to related data in other tables, enabling efficient querying and data integrity across the database.

QUESTION 742

How do conformed dimensions benefit data warehousing?

Answer

Conformed dimensions ensure that dimensions used across different fact tables are consistent in definition and structure, allowing for easier integration and analysis.

QUESTION 743

Define 'lift of a promotion'.

Answer

Lift of a promotion is the increase in sales over the baseline value that can be attributed to the effects of a promotional campaign.

QUESTION 744

Why is it important to document the capabilities of primary subsystems?

Answer

Documenting capabilities ensures that the design meets specific business needs and can guide implementation and evaluations.

Flashcards 745 - 756

QUESTION 745

What are the key attributes of the 'service line dimension'?

Answer

The service line dimension includes identifiers and descriptions related to specific services offered, crucial for analyzing billing and customer service data.

QUESTION 746

How can inventory levels impact retail profitability?

Answer

Optimized inventory levels reduce out-of-stocks and carrying costs, directly affecting sales and overall profitability for retailers.

QUESTION 747

What is the significance of having multiple roles for dimensions in a health care billing system?

Answer

Multiple roles for dimensions allow for different perspectives on the same data, such as distinguishing between an admitting provider and an attending provider.

QUESTION 748

What are the common pitfalls in building a data warehouse?

Answer

Common pitfalls include focusing too much on technology rather than business needs, neglecting user acceptance, and creating overly complex data structures.

QUESTION 749

What is the role of OLAP in financial analysis?

Answer

Online Analytical Processing (OLAP) provides tools for analyzing financial data, allowing users to perform complex queries and generate reports efficiently, enhancing decision-making capabilities.

QUESTION 750

Describe the role of the bridge table in reporting revenue.

Answer

The bridge table allows analysis of revenue at any hierarchical level by connecting the customer dimension and fact tables, enabling queries that consider the hierarchical structure of customers.

QUESTION 751

What considerations are crucial for the deployment of analytic applications?

Answer

Deployment should include a user acceptance testing phase, ensuring that applications meet business needs and that users are trained and supported in using the new tools.

QUESTION 752

What can the session dimension tell us about customer behavior?

Answer

The session dimension can indicate how many customers consult product information before ordering and track their actions during the session.

QUESTION 753

Explain the purpose of order management in a business.

Answer

Order management involves overseeing the order fulfillment process, from order placement to delivery, ensuring timely and accurate delivery of products to customers.

QUESTION 754

What are periodic snapshot models in inventory management?

Answer

Periodic snapshot models capture inventory levels at regular intervals, allowing analysis of inventory changes over time.

QUESTION 755

What should be included in the technical architecture documentation?

Answer

The documentation should include implementation phases, detailed design specifications, and a communication plan for stakeholders.

QUESTION 756

How do survey instruments contribute to HR data analysis?

Answer

Survey instruments collect feedback from employees, allowing HR to analyze performance ratings and make informed decisions about employee development and satisfaction.

Flashcards 757 - 768

QUESTION 757

What are accumulating snapshots in a healthcare data model?

Answer

Accumulating snapshots are fact tables that capture ongoing measurements over time, such as billing line items for healthcare services.

QUESTION 758

What is the significance of the most recent transaction indicator?

Answer

The most recent transaction indicator allows quick retrieval of the latest employee profile, ensuring accurate and timely reporting.

QUESTION 759

What additional facts can be included in an enhanced inventory snapshot?

Answer

Enhanced snapshots may include quantity sold, dollar value at cost, and dollar value at the latest selling price to analyze inventory performance more comprehensively.

QUESTION 760

What is a Temporary Price Reduction (TPR)?

Answer

A Temporary Price Reduction (TPR) is a promotional technique used in retail environments to temporarily lower prices.

QUESTION 761

What is a unit-of-measure conversion factor?

Answer

A unit-of-measure conversion factor is a multiplier or divisor used to convert quantities from one unit to another, such as converting order quantities to shipment quantities.

QUESTION 762

What organizational challenges may arise during the allocation process?

Answer

Allocation often involves political negotiations among departments, particularly when the finance department is responsible for establishing allocation rules.

QUESTION 763

What is the type 2 approach in SCD?

Answer

Type 2 creates a new row in the dimension table for each change, preserving historical data while allowing for current attribute values.

QUESTION 764

What are the unique characteristics of clickstream data?

Answer

Clickstream data includes dimensions such as page, session, and referrer, and presents challenges in identifying visitors and their behavior over sessions.

QUESTION 765

What is the role of a banner-ad server?

Answer

A banner-ad server determines which ad a visitor should receive based on their interests or previous interactions, logs the ad placed, and may send a cookie for future reference.

QUESTION 766

What is the importance of surrogate keys in dimensional modeling?

Answer

Surrogate keys are systematically generated unique identifiers used in place of operational keys to ensure consistency and efficiency in data management.

QUESTION 767

What is the purpose of dimension normalization?

Answer

Dimension normalization aims to reduce redundancy and improve data integrity by organizing data in a way that minimizes duplicate information.

QUESTION 768

What is a factless fact table?

Answer

A factless fact table is a table that captures events or conditions without associated measures, often used to track occurrences like attendance or student registrations.

Flashcards 769 - 780

QUESTION 769

What is the role of the POS transaction number in the retail sales schema?

Answer

The POS transaction number serves as a unique identifier for transactions, facilitating grouping of line items in the fact table, even without a corresponding dimension table.

QUESTION 770

What is the significance of the product dimension in retail sales?

Answer

The product dimension describes every SKU in the store, containing essential attributes that enable detailed analysis of sales across different merchandise hierarchies.

QUESTION 771

What does inventory turnover indicate?

Answer

Inventory turnover indicates how quickly inventory is sold and replaced over a given period, reflecting the efficiency of inventory management.

QUESTION 772

What are the implications of mixing different fact granularities?

Answer

Mixing fact granularities (e.g., order and order line facts) within a single fact table can lead to confusion and inaccurate analysis; separate tables or allocations should be used.

QUESTION 773

How does the Internet appliance affect data warehousing services?

Answer

Internet appliances enhance data warehousing by enabling real-time data collection through user interactions and facilitating various services such as e-learning, shopping, and information retrieval.

QUESTION 774

Why should to-date columns be avoided in fact tables?

Answer

To-date columns can lead to nonsensical and overstated results when fact rows are queried and summarized, as they do not align with the grain of the data.

QUESTION 775

What are the characteristics of effective interviewing?

Answer

Effective interviewing involves active listening, establishing rapport with the interviewee, verifying communication, and capturing precise terminology.

QUESTION 776

What are some challenges in tracking clickstream data?

Answer

Challenges include identifying visitor origins, sessions, and identities, as well as issues with browser caches and proxy servers.

QUESTION 777

Define the event dimension in clickstream analysis.

Answer

The event dimension describes actions taken on a page, such as opening a page, refreshing, clicking links, or entering data.

QUESTION 778

What techniques can be used to select products for the data warehouse?

Answer

Techniques include developing a product evaluation matrix based on architectural requirements, conducting market research, and understanding corporate purchasing processes.

QUESTION 779

What is temporal inconsistency?

Answer

Temporal inconsistency refers to the tendency of an OLTP database to change its primary data relationships frequently, impacting users by presenting constantly changing data and potentially losing historical data.

QUESTION 780

What are the characteristics of dirty dimensions?

Answer

Dirty dimensions refer to dimensions that are difficult to reliably identify and track across different claims, often due to the involvement of multiple individuals or entities.

Flashcards 781 - 792

QUESTION 781

What is an Implementation Bus Matrix?

Answer

A detailed version of the data warehouse bus matrix identifying fact tables for each business process along with their granularity and measurements.

QUESTION 782

Why might a designer choose to create a junk dimension?

Answer

A junk dimension consolidates low-cardinality flags and indicators into a single dimension, preventing fact table bloating and simplifying the data model.

QUESTION 783

Define Superdimension in the context of dimensional modeling.

Answer

A superdimension combines multiple related dimensions into one for simplification, typically used when the individual dimensions have small row counts.

QUESTION 784

What is the significance of dimensional modeling in data warehousing?

Answer

Dimensional modeling simplifies data structures to improve user comprehension and query performance, aligning with the needs of business users for intuitive access to data.

QUESTION 785

Why is it essential to involve finance or business teams in the allocation effort?

Answer

Involving finance or business teams ensures that allocation rules are established based on operational realities, preventing delays for the data warehouse team.

QUESTION 786

What is a degenerate dimension in a fact table?

Answer

A degenerate dimension is a dimension key without a corresponding dimension table, often representing identifiers like transaction numbers that are useful for grouping data but lack additional descriptive attributes.

QUESTION 787

What are some core principles of Customer Relationship Management (CRM)?

Answer

CRM focuses on managing a company's interactions with current and potential customers through data analysis, improving relationships, and enhancing customer satisfaction and retention.

QUESTION 788

Define the chart of accounts.

Answer

The chart of accounts is a structured list of all accounts used in the general ledger, categorized by account type (assets, liabilities, income, etc.) and often containing hierarchical relationships for reporting.

QUESTION 789

What is the importance of defining success criteria during interviews?

Answer

Defining success criteria allows interviewees to articulate measurable expectations for the project, ensuring alignment between user needs and deliverables.

QUESTION 790

What are some key reasons for maintaining separate dimensions rather than combining them?

Answer

Separate dimensions are required when data volumes are large, or when different business processes need unique attributes that do not conform to combined dimensions.

QUESTION 791

What is the value of a Terabyte (TB)?

Answer

A Terabyte (TB) is equivalent to one trillion bytes (10^12 bytes).

QUESTION 792

What dimensions are included in the clickstream fact table for complete sessions?

Answer

The dimensions included are calendar date, time of day, customer, page, session, and referrer. These dimensions help in analyzing session-related metrics.

Flashcards 793 - 804

QUESTION 793

How can the combining of dimensions affect reporting?

Answer

Combining dimensions can simplify reporting by providing a unified view of related attributes, but it may also limit flexibility if data volumes increase.

QUESTION 794

What is a factless fact table?

Answer

A factless fact table records many-to-many correlations, such as the relationships between loss parties and loss items in accident claims, without any measurable facts.

QUESTION 795

What is the relationship between sales organization changes and data reporting?

Answer

Changes in sales organization structures may require adjustments in how data is reported to accommodate historical and current organizational views.

QUESTION 796

What characterizes an Accumulating Snapshot Fact Table?

Answer

An Accumulating Snapshot Fact Table represents the full life of a transaction or discrete product, with multiple date stamps indicating major events, and is updated frequently.

QUESTION 797

What is 'data feasibility' in the context of warehouse project readiness?

Answer

Data feasibility assesses whether real data is being collected in operational systems to support business requirements, critical for project success.

QUESTION 798

What are some typical numeric facts in retail sales?

Answer

Typical numeric facts include sales quantity, sales dollar amount, cost dollar amount, and gross profit.

QUESTION 799

How do operational data stores (ODS) fit into data warehousing?

Answer

ODS are updated copies of operational data used for operational reporting and real-time interactions, serving as a bridge between operational systems and the data warehouse.

QUESTION 800

What are the attributes included in the HR periodic snapshot with an audit dimension?

Answer

Attributes include extract completion date and time, extract status, number of records extracted, transformation status, load status, and various employee metrics like salary paid and vacation days accrued.

QUESTION 801

What is the importance of granularity in fact tables?

Answer

Granularity refers to the level of detail captured in fact tables. Each different measurement grain requires its own fact table to avoid overcounting and ensure accurate data representation.

QUESTION 802

What is an accumulating snapshot model?

Answer

An accumulating snapshot model records a single row for each product delivery and updates it as the product moves through the inventory stages.

QUESTION 803

What is the purpose of the employee transaction dimension?

Answer

The employee transaction dimension captures and tracks changes to employee profiles, allowing for detailed analysis of employee transactions.

QUESTION 804

What are the primary components of a data warehouse?

Answer

The primary components of a data warehouse include data sources, ETL (Extract, Transform, Load) processes, a central repository, and access tools for reporting and analysis.

Flashcards 805 - 816

QUESTION 805

How do dimension tables facilitate reporting?

Answer

Dimension tables provide the descriptive attributes needed for report labels and groupings, making data more accessible and understandable for users.

QUESTION 806

What is a multivalued dimension?

Answer

A multivalued dimension occurs when an account can have multiple individual account holders or customers associated with it, requiring the use of a bridge table to maintain granularity.

QUESTION 807

What is included in a profit and loss (P&L) statement?

Answer

A P&L statement includes revenues and costs arranged from top line (undiscounted value) to bottom line (contribution), providing insights into financial performance.

QUESTION 808

What is the importance of capturing demographic information in a financial services context?

Answer

Demographic information helps banks understand their customer base, tailor products, and improve marketing strategies, enhancing customer engagement and service.

QUESTION 809

What is a key factor in the success of a data warehouse?

Answer

The active involvement of business users in the design and implementation phases ensures that the data warehouse aligns with their needs and promotes user adoption.

QUESTION 810

What is the grain of a procurement transaction fact table?

Answer

The grain of a procurement transaction fact table is defined as one row per procurement transaction, with key dimensions including transaction date, product, vendor, contract terms, and procurement transaction type.

QUESTION 811

What role does the ICD-9-CM standard play in health care?

Answer

The ICD-9-CM standard provides structured codes for diagnosing and treating medical conditions, facilitating billing and data analysis.

QUESTION 812

What factors should be considered when deciding between multiple versus single transaction fact tables?

Answer

Consider users' analytic requirements, whether there are distinct business processes, the involvement of multiple source systems, and the dimensionality of the facts. Separate fact tables are often favored when processes are distinct and data comes from different sources.

QUESTION 813

What is metadata in the context of a data warehouse?

Answer

Metadata is information that describes other data in the data warehouse, including definitions, schemas, and rules for data transformation and loading, serving as a guide for data management.

QUESTION 814

What is dimension role-playing?

Answer

Dimension role-playing occurs when a single dimension is used multiple times in a fact table, serving different roles simultaneously, typically achieved through views.

QUESTION 815

What is the role of the promotion dimension in retail sales?

Answer

The promotion dimension tracks details about promotions, such as names and dates, and links to sales data for analysis of promotional effectiveness.

QUESTION 816

What are the four steps in designing dimensional models?

Answer

1. Select the business process to model, 2. Declare the grain of the business process, 3. Choose the dimensions for each fact table row, 4. Identify the numeric facts for each fact table row.

Flashcards 817 - 828

QUESTION 817

What is the purpose of monitoring commitments in budgeting?

Answer

Managers monitor commitments to compare them with the annual budget in order to manage spending effectively.

QUESTION 818

Explain the difference between fact and dimension attributes.

Answer

Fact attributes are numeric measures used for analysis, while dimension attributes provide context and descriptions for those facts.

QUESTION 819

What is the significance of the insurance dimension in data warehousing?

Answer

The insurance dimension captures vital attributes related to policyholders, claims, and coverage details, essential for risk assessment and financial analysis.

QUESTION 820

What are the three types of change responses in dimensional modeling?

Answer

The three types of change responses are Type 1 (overwrite the attribute value), Type 2 (add a new row with a new surrogate key), and Type 3 (modify existing row attributes without adding a new row).

QUESTION 821

What is the role of security policies in data warehouses?

Answer

Security policies dictate how personal information is managed, accessed, and protected within a data warehouse, ensuring compliance with legal and ethical standards.

QUESTION 822

How do business users view the importance of data in insurance companies?

Answer

Business users see data as a critical asset for identifying opportunities and responding effectively to market changes.

QUESTION 823

What are the expected future trends in data warehousing?

Answer

Future trends may include increased use of artificial intelligence for data processing, enhanced privacy controls, and the integration of diverse data sources for real-time analytics.

QUESTION 824

Why is it essential to understand the grain in fact tables?

Answer

Understanding the grain ensures that data is accurately represented and prevents issues with summarization and reporting, particularly for semi-additive facts.

QUESTION 825

What is the purpose of documenting the technical architecture?

Answer

Documenting the technical architecture provides a clear plan for implementation, ensuring all team members understand the framework and can proceed accordingly.

QUESTION 826

What is the chart of accounts?

Answer

The chart of accounts is a structured list of all account identifiers in the G/L, categorizing accounts into types like assets, liabilities, and expenses.

QUESTION 827

Why is it important for CRM to have a strategy and plan?

Answer

A proper strategy and plan ensure that the CRM implementation aligns with business objectives and effectively integrates technology with organizational processes.

QUESTION 828

Why is it important to avoid null keys in a fact table?

Answer

Avoiding null keys in a fact table maintains referential integrity and prevents confusion for users when performing joins or queries on the data.

Flashcards 829 - 840

QUESTION 829

What is a 'dimension role-playing technique'?

Answer

Dimension role-playing allows a single dimension to be used in multiple contexts within the same schema, such as using the date dimension to represent both order date and ship date.

QUESTION 830

How does the customer dimension contribute to a clickstream data mart?

Answer

The customer dimension provides essential information about the customers, allowing for segmentation and analysis of customer behaviors and preferences across sessions.

QUESTION 831

What challenges do privacy laws pose for data warehouses?

Answer

Privacy laws require data warehouses to implement strict data management practices, potentially complicating data access and usage while ensuring compliance.

QUESTION 832

What is the purpose of the requirements wrap-up presentation?

Answer

The requirements wrap-up presentation reviews findings with stakeholders, facilitates consensus on priorities, and helps establish a roadmap for future data warehouse development.

QUESTION 833

What are additive facts?

Answer

Additive facts are numerical measures that can be summed across any dimension in a fact table, such as total sales or quantity sold.

QUESTION 834

What are conformed dimensions and why are they important?

Answer

Conformed dimensions are shared dimensions across multiple data marts, ensuring consistency and enabling integration between different data marts within a data warehouse.

QUESTION 835

How might historical data influence future financial planning?

Answer

Historical data provides insights into past performance and trends, which can inform future budgeting and financial strategies.

QUESTION 836

Define 'control numbers' in the context of procurement.

Answer

Control numbers are unique identifiers created during various procurement processes, such as purchase order numbers and payment check numbers, essential for tracking transactions.

QUESTION 837

What is the purpose of a data warehouse bus matrix?

Answer

It helps to visualize the relationships between business processes and dimensions, promoting consistency and integration across data marts.

QUESTION 838

What are catastrophic failures in data warehousing?

Answer

Catastrophic failures refer to significant disruptions in data warehouse operations, which can be caused by physical destruction, insider sabotage, cyberattacks, or single-point failures.

QUESTION 839

How do dimension tables enhance data analysis in a data warehouse?

Answer

Dimension tables enhance data analysis by providing descriptive attributes that enable detailed filtering, grouping, and labeling of data, facilitating comprehensive reporting and insights.

QUESTION 840

What is the role of a data access tool in analyzing customer behavior?

Answer

Data access tools simplify the process of querying complex customer behavior analyses, allowing users to extract insights without needing extensive SQL knowledge.

Flashcards 841 - 852

QUESTION 841

What is the significance of a communication matrix in project management?

Answer

A communication matrix outlines how and when to communicate with different stakeholders, ensuring consistent updates and alignment throughout the project.

QUESTION 842

How does telecommunications impact daily life?

Answer

Telecommunications enables communication over distances, affecting how people connect, access information, and engage with services in both personal and professional contexts.

QUESTION 843

What is the significance of effective and expiration dates in sales rep coverage?

Answer

Effective and expiration dates in sales rep coverage help track historical assignments, ensuring that analyses reflect current and past relationships between sales reps and customers.

QUESTION 844

What is the function of a dimension table?

Answer

Dimension tables provide descriptive attributes related to the facts in a fact table, facilitating detailed analysis.

QUESTION 845

Define Accumulating Snapshot.

Answer

A fact table that maintains a single row representing the entire history of a process, often used for tracking items through various stages.

QUESTION 846

What is the role of a dimension authority?

Answer

The dimension authority is responsible for defining, maintaining, and publishing the conformed dimensions to all data mart clients, ensuring high-quality and consistent dimension tables across the enterprise.

QUESTION 847

What are the benefits of integrating the clickstream data mart into the enterprise data warehouse?

Answer

Integrating the clickstream data mart allows for comprehensive analysis across various business processes, facilitating insights into customer behavior and enhancing decision-making.

QUESTION 848

What is a Surrogate Key?

Answer

Integer keys assigned in the staging area to populate a dimension table and join it to a fact table, often lacking inherent meaning.

QUESTION 849

Define Liability in accounting.

Answer

An item on a balance sheet representing money owed by a company to others.

QUESTION 850

Explain the concept of a companion transaction grained fact table.

Answer

A companion transaction grained fact table may be used to capture detailed information about specific transactions when accumulating snapshots are insufficient.

QUESTION 851

What is the purpose of financial services in the context of data warehousing?

Answer

Financial services leverage data warehousing for analyzing financial transactions, risk management, compliance, and strategic planning, enhancing decision-making.

QUESTION 852

What are the primary components to design and specify in a data warehouse subsystem?

Answer

The primary components include data staging services, security requirements, physical infrastructure, and configuration needs.

Flashcards 853 - 864

QUESTION 853

What are dimension tables?

Answer

Dimension tables contain descriptive attributes related to the dimensions of the business, serving as the primary source of query constraints and report labels in a data warehouse.

QUESTION 854

What are the primary components of a Billing Fact Table?

Answer

The Billing Fact Table typically includes keys linking to customer, service line, sales rep, and billing date dimensions, along with quantitative metrics like service charges and taxes.

QUESTION 855

What are surrogate keys?

Answer

Surrogate keys are integers assigned sequentially as needed to populate a dimension. They serve to join dimension tables to the fact table without embedding intelligence in the keys.

QUESTION 856

What is the GROUP BY clause in SQL?

Answer

An SQL clause that uniquely lists the unaggregated items in the SELECT list, excluding those involved in SUM, COUNT, MIN, MAX, or AVG.

QUESTION 857

What is the role of accounting in a business?

Answer

Accounting involves the systematic recording, reporting, and analysis of financial transactions, providing stakeholders with insights into the company's financial health.

QUESTION 858

How does the dimensional design affect data storage?

Answer

A streamlined dimensional design leads to more efficient data storage, reducing the size of fact tables and indices, which enhances performance.

QUESTION 859

Explain the significance of independent hierarchies in a customer ship-to dimension.

Answer

Independent hierarchies allow for multiple organizational and geographic breakdowns, such as by city, county, and state, enhancing data analysis and reporting capabilities.

QUESTION 860

What does the term 'snowflaking' refer to in dimensional modeling?

Answer

Snowflaking is the process of normalizing dimension tables by creating secondary tables to reduce redundancy, though it can complicate schema and slow performance.

QUESTION 861

What is the purpose of the findings document after gathering requirements?

Answer

The findings document organizes insights from interviews, outlines key business processes, and serves as a basis for prioritizing data warehouse projects.

QUESTION 862

Define the term 'fact table'.

Answer

A fact table is a central table in a star schema of a data warehouse that stores quantitative data for analysis, typically consisting of measurements and foreign keys to dimension tables.

QUESTION 863

What is the importance of fiscal responsibility in an organization?

Answer

Fiscal responsibility ensures that financial resources are used effectively, promoting accountability and sustainable financial practices.

QUESTION 864

Define accumulating snapshot grain.

Answer

The accumulating snapshot grain represents a single row in a fact table that captures the cumulative history of a measurement over time, often used for tracking processes like billing.

Flashcards 865 - 876

QUESTION 865

What is the significance of automated procedures in ensuring data quality?

Answer

Automated procedures help detect errors and inconsistencies in data, ensuring that the data warehouse maintains high standards of data integrity and accuracy.

QUESTION 866

How do transaction-grained fact tables differ from snapshot fact tables?

Answer

Transaction-grained fact tables focus on individual transactions with specific metrics, while snapshot fact tables capture data at specific points in time, aggregating metrics over periods.

QUESTION 867

Explain the concept of 'outrigger' in dimensional modeling.

Answer

An outrigger is a secondary dimension that extends the primary dimension's attributes, often used to provide additional context without cluttering the main dimension.

QUESTION 868

What is the implication of using type 2 SCD with large dimensions?

Answer

Using type 2 SCD with large dimensions can lead to performance issues and complexity, prompting the use of techniques like minidimensions or bridge tables to manage changes.

QUESTION 869

What are the goals of a retail bank as discussed in the context?

Answer

The goals include analyzing accounts, marketing effectively to existing customers, and understanding the household relationships to enhance product offerings.

QUESTION 870

What is a cookie in the context of web interactions?

Answer

A cookie is a small piece of data stored on the user's computer by the web browser that can be used to track user behavior and preferences.

QUESTION 871

What is an SKU?

Answer

An SKU (Stock Keeping Unit) is a unique identifier for each product variant sold, including those without universal product codes (UPCs).

QUESTION 872

What does the term 'page seconds' refer to?

Answer

'Page seconds' measures the time elapsed before the next page event during a customer's session, contrasting with 'session seconds' which measures total session duration.

QUESTION 873

Explain the concept of 'factless fact tables.'

Answer

Factless fact tables are used to capture many-to-many relationships without numerical measures, often used to track events or conditions.

QUESTION 874

How does the time-stamped transaction tracking work in HR data?

Answer

Each employee transaction event is captured in a transaction-grained fact table, recording changes made to employee profiles over time.

QUESTION 875

How do we determine the success status of a session?

Answer

The success status is evaluated based on whether the overall mission of the session, such as making a purchase, was achieved. It requires analyzing the complete session data at the time of data extraction.

QUESTION 876

Define 'fact table' in dimensional modeling.

Answer

A fact table is a central table in a star schema that contains quantitative data for analysis, typically aggregated measures related to business processes.

Flashcards 877 - 888

QUESTION 877

How does the concept of 'watching the watchers' affect data privacy?

Answer

The concept suggests that increasing transparency in data gathering and usage can help mitigate privacy concerns and prevent abuses of personal information.

QUESTION 878

Why is it important to track costs in the profitability data mart?

Answer

Tracking costs allows businesses to accurately measure profitability and analyze the financial performance of various sales channels.

QUESTION 879

How do dimensional models handle varying product attributes?

Answer

Dimensional models utilize custom schemas or outriggers to capture varying attributes specific to each product type while maintaining a core fact table for common metrics.

QUESTION 880

What is the importance of dimension tables in a data warehouse?

Answer

Dimension tables provide descriptive attributes that allow users to analyze facts across various contexts, enhancing the usability of the data warehouse.

QUESTION 881

What is the significance of intellectual property rights in data warehousing?

Answer

Organizations must be cautious about collecting information from outside sources due to ownership rights, as misappropriation can lead to legal issues, especially with copyrighted works.

QUESTION 882

Define operational CRM.

Answer

Operational CRM refers to the systems and processes that manage customer interactions and data throughout the customer lifecycle, enhancing customer service and satisfaction.

QUESTION 883

What is the Type 2 approach for slowly changing dimensions?

Answer

The Type 2 approach creates a new dimension row with a new surrogate key for any attribute that needs to track historical changes, preserving the old data.

QUESTION 884

What considerations should be taken when modeling time of day?

Answer

Time of day can be modeled as a separate dimension or fact based on the need to roll up time periods for analysis.

QUESTION 885

What is a degenerate dimension?

Answer

A degenerate dimension is a dimension that does not have its own dimension table but is represented as an attribute in the fact table, such as a transaction ID.

QUESTION 886

What are the implications of high-bandwidth communications for data warehousing?

Answer

High-bandwidth communications enable faster data transfer and real-time analytics, allowing data warehouses to operate more efficiently and support more complex applications.

QUESTION 887

What does the term 'degenerate dimension' mean?

Answer

A degenerate dimension is an attribute that exists in a fact table without a corresponding dimension table, typically used for transactional numbers like invoice or order numbers.

QUESTION 888

What are permissible snowflakes?

Answer

Permissible snowflakes are exceptions to the general rule against snowflaking in data warehouses, where certain dimensional designs, like outriggers, may be justified.

Flashcards 889 - 900

QUESTION 889

What is the role of the band definition table in reporting?

Answer

The band definition table facilitates flexible value-band reporting by allowing multiple sets of different reporting bands to be defined and utilized in queries.

QUESTION 890

What are the challenges of implementing conformed dimensions?

Answer

Implementing conformed dimensions requires organizational consensus, technical coordination, and can face political challenges. It necessitates maintaining consistency and managing changes across multiple data marts.

QUESTION 891

What is a fact table?

Answer

A fact table is the primary table in a dimensional model that stores numerical performance measurements of the business, with each row representing a specific measurement at the intersection of different dimensions.

QUESTION 892

What are the challenges of using normalized models in data warehousing?

Answer

Normalized models can complicate data queries, making them difficult for users to understand and navigate, which defeats the purpose of simplifying data retrieval in a data warehouse.

QUESTION 893

What are the responsibilities of the support team post-deployment?

Answer

The support team must proactively assist users, resolve issues, and gather feedback to improve the data warehouse and maintain user engagement.

QUESTION 894

What is a slowly changing dimension (SCD)?

Answer

A slowly changing dimension is a data structure that manages and tracks changes to dimension attributes over time, ensuring historical accuracy.

QUESTION 895

Why should codes be minimized in dimension tables?

Answer

Minimizing codes in dimension tables enhances readability and usability, allowing users to understand data without relying on external code translations.

QUESTION 896

Define 'Third Normal Form' (3NF) in database design.

Answer

Third Normal Form (3NF) is a database design approach that eliminates redundancy and allows for the insertion of new rows without excessive data locking issues.

QUESTION 897

Discuss the significance of web visitor clickstream data.

Answer

Web visitor clickstream data provides insights into customer behavior online, allowing businesses to analyze interactions and improve marketing strategies.

QUESTION 898

What challenges may arise with multiple fact tables?

Answer

Managing and administering multiple fact tables can be time-consuming due to the need for additional loading, indexing, and aggregation.

QUESTION 899

What is allocated inventory?

Answer

Allocated inventory refers to stock that has been designated for shipment to a specific customer or store, even before the actual shipping occurs.

QUESTION 900

What is the importance of the session dimension in clickstream analysis?

Answer

The session dimension provides insights into visitor behavior, helping to analyze actions such as product consultations and order completions.

Flashcards 901 - 912

QUESTION 901

What are common dimensions in an insurance data mart?

Answer

Common dimensions include date, policyholder, coverage, covered item, and employee.

QUESTION 902

How can organizations analyze current budgeted amounts?

Answer

Organizations can constrain on all dates up to the present and sum amounts by department and line item to analyze current budgeted amounts.

QUESTION 903

What dimensions are associated with the policy transaction business process?

Answer

Dimensions include transaction date, effective date, policyholder, employee, coverage, covered item, policy number, and policy transaction type.

QUESTION 904

What is the role of a pricing actuary in the insurance policy process?

Answer

The pricing actuary determines the premium rate that will be charged for a policy based on specific coverages, covered items, and the qualifications of the policyholder.

QUESTION 905

What role do business executives play in the requirements gathering process?

Answer

Business executives provide vision and strategic direction, ensuring that the data warehouse deliverables align with organizational goals and needs.

QUESTION 906

Define the term 'fact table' in the context of data warehousing.

Answer

A fact table is a central table in a star schema of a data warehouse that stores quantitative data for analysis and is often denormalized.

QUESTION 907

What is a multivalued dimension attribute?

Answer

A multivalued dimension attribute allows for multiple values to be associated with a single dimension entry, such as multiple insured drivers for a policyholder.

QUESTION 908

What challenges arise when summarizing data by trip level in airline analysis?

Answer

Challenges include accurately identifying trip start and end points and determining applicable dimensions, as certain attributes may not apply at the trip level.

QUESTION 909

What are the benefits of using a centralized personal information database?

Answer

A centralized database provides a consistent and accurate view of personal data, simplifies data management, and enhances security through controlled access.

QUESTION 910

What are the advantages of keeping promotional conditions in one dimension?

Answer

Combining promotional conditions into one dimension simplifies browsing and analysis, allowing users to see the interactions between different promotional elements effectively.

QUESTION 911

What are the components of the POS retail sales transaction fact?

Answer

Components include date key, product key, store key, promotion key, POS transaction number, sales quantity, and various dollar amounts.

QUESTION 912

What is a 'fact dimension'?

Answer

A fact dimension is a special dimension used to identify extremely sparse or dissimilar measurements within a single fact table.

Flashcards 913 - 924

QUESTION 913

Describe the periodic snapshot in G/L data.

Answer

The periodic snapshot captures G/L account data at the end of each fiscal period, providing a single row for each accounting period.

QUESTION 914

What are operational and analytic roles of CRM?

Answer

Operational CRM focuses on synchronizing customer-facing processes across sales, marketing, and service. Analytic CRM leverages data collected from customer interactions to analyze behavior and improve strategies.

QUESTION 915

What are the keys in a custom product dimension?

Answer

The keys of custom product dimensions are the same as those used in the core product dimension, allowing for consistent integration across different product types.

QUESTION 916

What are the key metrics tracked in claims processing?

Answer

Key metrics include the original reserve amount, assessed damage amount, reserve adjustment amount, claim amount paid, and the number of transactions.

QUESTION 917

Why is understanding business requirements essential in data warehousing?

Answer

Understanding business requirements ensures that the data warehouse aligns with user needs, promotes buy-in, and supports effective decision-making.

QUESTION 918

What is a data warehouse, and how does it benefit financial data management?

Answer

A data warehouse provides a single source of usable, understandable financial information, ensuring consistency and clarity across the organization.

QUESTION 919

Describe the difference between 'leg' and 'segment' in airline data.

Answer

A leg is a flight from one airport to another without stops, while a segment can include multiple legs if the journey involves changes in aircraft.

QUESTION 920

How does the grain affect the choice of dimensions?

Answer

The grain determines the primary dimensions of the fact table, influencing what additional dimensions can be added without violating the grain.

QUESTION 921

What is the significance of demographic attributes in the household dimension?

Answer

Demographic attributes provide insights into household characteristics such as income, homeownership, and presence of children, helping banks tailor their services.

QUESTION 922

What is the difference between dimensional modeling and third-normal-form (3NF) modeling?

Answer

Dimensional modeling focuses on simplifying data for user understanding and efficient querying, while 3NF modeling aims to eliminate data redundancy by dividing data into many discrete entities, often resulting in complex structures.

QUESTION 923

What is the challenge of clickstream data integration into a data warehouse?

Answer

Integrating clickstream data into a data warehouse involves conforming its dimensions and facts to align with existing data structures.

QUESTION 924

What are the consequences of having poor data quality in a data warehouse?

Answer

Poor data quality can lead to inaccurate analyses, misguided business decisions, and a loss of trust in the data warehouse.

Flashcards 925 - 936

QUESTION 925

What is the purpose of using an audit dimension?

Answer

An audit dimension can capture data lineage information for fact rows, including the time of the extract and the source table, providing traceability of data.

QUESTION 926

What is the main focus of the data warehouse lifecycle?

Answer

The data warehouse lifecycle encompasses the entire process of data warehouse design, implementation, and maintenance, emphasizing the need for integrated activities and ongoing communication.

QUESTION 927

How can a visitor's session be identified in a clickstream?

Answer

A visitor's session can be identified using a unique session ID, which can be established through cookies, time-contiguous log entries, or other tracking methods.

QUESTION 928

What are the common types of fact tables based on granularity?

Answer

Fact tables can be categorized into transaction, periodic snapshot, and accumulating snapshot based on the granularity of the data they contain.

QUESTION 929

What is the potential future direction of data warehousing?

Answer

Future directions may include greater integration with real-time data processing, enhanced data visualization tools, and a stronger emphasis on security and privacy measures.

QUESTION 930

What is the purpose of a glossary in a data warehousing context?

Answer

A glossary serves as a reference resource, helping users understand key terms and concepts related to data warehousing and dimensional modeling.

QUESTION 931

What are common data warehousing pitfalls to avoid?

Answer

Common pitfalls include neglecting user engagement, failing to prioritize actionable feedback, and underestimating the need for ongoing support and education post-deployment.

QUESTION 932

What are the key implications of the information revolution for data warehousing?

Answer

The information revolution requires constant adaptation of data warehousing practices due to rapid technological changes, user expectations, and the evolving information environment.

QUESTION 933

What role does the date dimension play in inventory management?

Answer

The date dimension allows for tracking inventory levels over time, facilitating temporal analysis of stock availability and trends.

QUESTION 934

How does a coverage table differ from a traditional fact table?

Answer

A coverage table records events or situations (like facility usage) without numeric measurements, while traditional fact tables include measurable facts.

QUESTION 935

What is the significance of dimension attributes like package type in the product dimension?

Answer

Dimension attributes such as package type provide additional filtering options for analysis, allowing users to group and compare products across different packaging formats.

QUESTION 936

What is a factless fact table?

Answer

A factless fact table is a table that records the relationship between dimensions without any measurable facts. It is often used to track events or coverage without associated numerical data.

Flashcards 937 - 948

QUESTION 937

What is a minidimension?

Answer

A minidimension is a smaller dimension table used to track rapidly changing attributes separately from the main dimension, linked to the fact table with a surrogate key.

QUESTION 938

What attributes should be included in an audit dimension?

Answer

Attributes may include extract completion dates, statuses for transformations and loads, and counts of records processed to maintain data quality and lineage.

QUESTION 939

Why is application integration critical in data warehousing?

Answer

Application integration is essential for achieving a consistent view of customer-facing processes, often requiring significant changes to operational systems to resolve data compatibility issues.

QUESTION 940

Define the term 'data warehouse architecture'.

Answer

Data warehouse architecture refers to the structure that defines how data is collected, stored, accessed, and managed within a data warehouse.

QUESTION 941

Why is it important to treat accounts and households as separate dimensions?

Answer

It allows for better management of data granularity and volatility, as households may change more frequently than accounts, preventing unnecessary complications in dimension management.

QUESTION 942

How can the AND/OR dilemma be resolved in SQL?

Answer

By using unions for OR queries and custom SQL code for AND queries, potentially employing intersections to manage constraints across rows.

QUESTION 943

What is subrogation in insurance?

Answer

Subrogation is the process by which an insurance company sells the rights to recoveries from pending lawsuits or counterclaims to specialists, allowing the company to obtain cash immediately and close its books on the claim.

QUESTION 944

How is the claims transaction schema structured?

Answer

The schema includes facts and dimensions related to claims transactions, such as claim details, policyholder information, and third-party involvement, ensuring accurate tracking of claims processing.

QUESTION 945

What is Meta Data?

Answer

Data maintained to support the operations or use of a data warehouse, akin to an encyclopedia for the data warehouse.

QUESTION 946

What happens if header-level facts cannot be allocated successfully?

Answer

If header-level facts cannot be allocated, they should be presented in aggregate form, limiting the ability to analyze them by product dimension.

QUESTION 947

What challenges arise with many-to-many relationships in ship-to and bill-to dimensions?

Answer

Many-to-many relationships complicate the hierarchy assumed in the customer ship-to dimension, requiring careful design choices to ensure data integrity and usability.

QUESTION 948

What are the implications of having multiple lines of business in an insurance company?

Answer

Different lines of business may require unique dimensions and fact tables, complicating data integration and analysis.

Flashcards 949 - 960

QUESTION 949

What is the Type 3 response and when is it used?

Answer

The Type 3 response adds a new column to capture the old value while overwriting the current value, allowing both to be visible for analysis.

QUESTION 950

What are the two approaches to handling commercial customer hierarchies?

Answer

The two approaches are fixed-depth hierarchies, which use distinct attributes for each level, and variable-depth hierarchies, which use a recursive parent key to represent complex relationships.

QUESTION 951

Define 'clickstream'.

Answer

Clickstream is the composite body of actions taken by a user at a web browser, including actual clicks and server responses, often recorded in web server logs.

QUESTION 952

What is the significance of the general ledger in accounting?

Answer

The general ledger acts as the core financial system that integrates detailed information from purchasing, payables, and receivables subledgers, facilitating comprehensive financial management.

QUESTION 953

What is the potential impact of GPS integration on data warehousing?

Answer

GPS integration allows for enhanced tracking and analytics of data based on geographic location, improving the relevance and utility of collected data.

QUESTION 954

How does an account-to-customer bridge table function?

Answer

The account-to-customer bridge table links individual customer dimensions to an account-grained fact table using surrogate account and customer foreign keys.

QUESTION 955

What are the three options when confronted with scope changes?

Answer

1. Increase the scope by adding time, resources, or budget. 2. Maintain total effort while relinquishing something previously in scope. 3. Handle the change as an enhancement request without officially agreeing to it.

QUESTION 956

What is the significance of the average daily balance in financial services?

Answer

The average daily balance provides insights into account usage and customer behavior over a specified period, supporting financial analysis and decision-making.

QUESTION 957

What are the benefits of using a SAN in data warehousing?

Answer

The benefits include the ability to locate disk drives and backup devices in separate buildings, perform disk-to-disk backup at high speeds, and share resources for multiple application systems.

QUESTION 958

What is a B-tree index?

Answer

A B-tree index is a relational index that organizes data in a tree structure, allowing for efficient retrieval, especially for high-cardinality columns.

QUESTION 959

What is a Profit-and-Loss (P&L) Schema?

Answer

An income statement representation of revenues and costs, illustrating the progression to net profit and allowing slicing and dicing of financial data.

QUESTION 960

What should follow the deployment of a data warehouse?

Answer

Post-deployment, ongoing user support, education, technical support, and program support are crucial to ensure continued engagement and effective use of the data warehouse.

Flashcards 961 - 972

QUESTION 961

What is the significance of using surrogate keys in the date dimension?

Answer

Surrogate keys serve as simple integer representations of dates, allowing for efficient partitioning and management of data while avoiding null values for special conditions.

QUESTION 962

What is the expected structure of a promotion dimension table?

Answer

A promotion dimension table typically includes columns for promotion key, promotion name, price reduction type, promotion media type, ad type, display type, coupon type, and cost.

QUESTION 963

How does the structure of a Web page affect clickstream data?

Answer

The structure, including static versus dynamic pages, influences how events are logged and how accurately visitor sessions can be tracked.

QUESTION 964

What is the role of an aggregate navigator?

Answer

An aggregate navigator is software that enhances query performance by transforming user SQL queries to utilize pre-calculated summary data (aggregates) in the data warehouse.

QUESTION 965

What is the significance of handling time zones in data warehousing?

Answer

To ensure accurate reporting of transaction dates and times across different geographical locations, accommodating local and standardized time representations.

QUESTION 966

What is the future outlook for data warehousing?

Answer

The future outlook includes advancements in technology, increased emphasis on privacy and security, and the continuous evolution of user expectations and data usage.

QUESTION 967

What is subrogation in insurance?

Answer

Subrogation is the process where an insurance company sells the rights to recoveries from claims to specialists, allowing them to close their books on a claim.

QUESTION 968

What does the customer ship-to dimension represent?

Answer

The customer ship-to dimension contains one row for each unique location to which products are shipped, including various geographic and organizational hierarchies.

QUESTION 969

What is the relationship between facts, dimensions, and the grain in a dimensional model?

Answer

Facts are the measurements analyzed, dimensions provide context to these measurements, and the grain defines the level of detail for both.

QUESTION 970

What are the implications of using a fact dimension for medical records?

Answer

Using a fact dimension allows for a flexible design that can accommodate the variability of medical records without creating excessive nulls.

QUESTION 971

What is the challenge of point-in-time balances?

Answer

Point-in-time balances require capturing account balances at arbitrary historical points, which can be overwhelming if daily snapshots are created for large numbers of accounts.

QUESTION 972

What is a 'superdimension' in dimensional modeling?

Answer

A superdimension combines multiple dimensions into a single entity to simplify analysis, particularly when dealing with small dimension tables.

Flashcards 973 - 984

QUESTION 973

What is the significance of tracking customer behavior in financial services?

Answer

Tracking customer behavior allows financial institutions to tailor their services, predict future needs, and improve customer satisfaction, thus enhancing profitability.

QUESTION 974

What is the importance of having detailed attributes in the product dimension?

Answer

Having detailed attributes in the product dimension enables robust analysis capabilities, allowing users to filter and drill down into specific product characteristics for better insights.

QUESTION 975

What is the purpose of the most recent surrogate key assignment table?

Answer

This table facilitates fast lookups when assigning surrogate keys to fact tables and keeps track of the most recent surrogate keys for operational source keys.

QUESTION 976

What is the purpose of dimension triage in dimensional modeling?

Answer

Dimension triage helps to avoid having too few dimensions by ensuring all relevant aspects of the business are captured appropriately, improving usability and performance of the model.

QUESTION 977

What is a household dimension in dimensional modeling?

Answer

The household dimension provides a smaller point of entry into the fact table without traversing a large account dimension table, capturing relationships between accounts and households.

QUESTION 978

What are some challenges associated with general medical records in a data warehouse?

Answer

General medical records are highly variable, consisting of different forms such as numeric data, unique lab results, free-text comments, and images.

QUESTION 979

Explain the concept of 'minidimensions' in data warehousing.

Answer

Minidimensions are subsets of more extensive dimension tables, used to handle attributes that can vary rapidly, such as customer preferences or sales rep performance.

QUESTION 980

What is a minidimension in the context of CRM?

Answer

Minidimensions are smaller, specialized tables that support attribute browsing and change tracking for larger dimensions, helping to manage complexity and improve performance.

QUESTION 981

What is a fact table?

Answer

A fact table is a central table in a star schema of a data warehouse that stores quantitative data for analysis and is often denormalized for performance reasons.

QUESTION 982

What is the role of fact tables?

Answer

Fact tables store quantitative data for analysis, containing measurable metrics that are often linked to dimension tables for context.

QUESTION 983

What is a 'factless fact table'?

Answer

A factless fact table is a table that does not contain any measurable facts but captures certain many-to-many relationships between dimension keys, often used to represent events.

QUESTION 984

What is dimensional modeling?

Answer

Dimensional modeling is a design technique used in data warehouses that structures data into fact and dimension tables to facilitate efficient querying and reporting.

Flashcards 985 - 996

QUESTION 985

Define a fact table.

Answer

A fact table is a central table in a star schema of a data warehouse that stores quantitative data for analysis and is often denormalized.

QUESTION 986

What is the significance of behavior study groups in CRM?

Answer

Behavior study groups track sets of customers who exhibit similar characteristics or behaviors, enabling targeted strategies and better understanding of customer needs.

QUESTION 987

What is the purpose of project planning in a data warehouse project?

Answer

Project planning assesses organizational readiness, establishes scope, and determines resources necessary for a successful data warehouse initiative.

QUESTION 988

What are the challenges of identifying a complete session in web analytics?

Answer

The stateless nature of HTTP makes it difficult to define sessions accurately, often requiring additional mechanisms like cookies.

QUESTION 989

What is the purpose of gathering business requirements?

Answer

Gathering business requirements is essential for understanding user needs and ensuring that the data warehouse aligns with business goals and provides value.

QUESTION 990

What is the role of OLAP in financial reporting?

Answer

OLAP products enable fast query performance for financial reporting, budgeting, and consolidation applications by using precalculated data.

QUESTION 991

What is Time Shifting of a promotion?

Answer

Time Shifting refers to the tendency of promotions to cause customers to delay purchases until the promotion is active, often leading to a decrease in purchases afterward.

QUESTION 992

What is the importance of Geographic Information Systems (GIS) in data warehousing?

Answer

GIS tools enhance data visualization and analysis by interpreting spatial data, allowing for richer insights into geographic trends and patterns.

QUESTION 993

What is a common mistake in dimensional modeling?

Answer

Failing to conform facts and dimensions across separate fact tables can lead to isolated data stovepipes and inaccurate comparisons of measured facts like revenue.

QUESTION 994

Explain the concept of conformed dimensions.

Answer

Conformed dimensions are shared dimensions across different fact tables, allowing for consistent reporting and analysis across multiple business processes.

QUESTION 995

What key dimensions are included in the retail sales schema?

Answer

Key dimensions in the retail sales schema include date, product, store, promotion, and POS transaction number.

QUESTION 996

What is a degenerate dimension?

Answer

A degenerate dimension is a dimension that does not have its own dimension table but is included in the fact table as an attribute, such as a control number that identifies a specific transaction.

Flashcards 997 - 1008

QUESTION 997

Define metadata in the context of data warehousing.

Answer

Metadata is data that describes other data, including definitions, calculations, and transformation rules, essential for understanding data lineage.

QUESTION 998

What is the natural granularity for an order transaction fact table?

Answer

The natural granularity for an order transaction fact table is one row for each line item on an order, allowing detailed tracking of individual items.

QUESTION 999

What is the significance of conformed dimensions?

Answer

Conformed dimensions are dimensions that are shared across multiple fact tables, ensuring consistency in reporting and analysis within a data warehouse.

QUESTION 1000

What are the advantages of using packet-filtering gateways in data warehouses?

Answer

Packet-filtering gateways enhance security by controlling access to sensitive data and ensuring that only trusted applications can communicate with the database.

QUESTION 1001

What is the role of a degenerate dimension in a fact table?

Answer

A degenerate dimension, such as an order number, is included in a fact table without a corresponding dimension table, allowing for grouping of line items while avoiding unnecessary joins.

QUESTION 1002

What are the challenges of integrating multiple operational order transaction systems?

Answer

Challenges include data inconsistency, urgency to integrate disparate results, and complexity in managing multiple source systems within a data warehouse.

QUESTION 1003

Explain the difference between policy transaction date and effective date.

Answer

The policy transaction date is when the transaction is entered into the system, while the effective date is when the transaction legally takes effect.

QUESTION 1004

What does 'Up-selling' mean?

Answer

Up-selling is a sales strategy where a seller encourages a customer to purchase a more expensive or higher-value version of a product they are already considering.

QUESTION 1005

What are the implications of CRM on data warehousing?

Answer

CRM requires data warehousing to support the integration of customer data from various sources, enabling comprehensive analysis and reporting on customer behavior and interactions.

QUESTION 1006

How do you calculate the number of rows in a bridge table for a customer organization?

Answer

To calculate the total number of rows, multiply the number of values at each hierarchical level by the depth of that level and sum the resulting products.

QUESTION 1007

What are surrogate keys?

Answer

Surrogate keys are unique identifiers assigned to dimension records that have no inherent meaning outside the database, often used to replace natural keys.

QUESTION 1008

What are the key functions of Human Resources Management?

Answer

Key functions of Human Resources Management include recruiting, training, performance management, employee relations, and compliance with labor laws.

Flashcards 1009 - 1020

QUESTION 1009

Define the term 'distributed architecture' in data warehousing.

Answer

Distributed architecture refers to a system design where data processing and storage are spread across multiple locations or servers, enhancing reliability and performance.

QUESTION 1010

What is the role of a project manager in a data warehouse initiative?

Answer

The project manager oversees planning, execution, and communication, ensuring that the project stays on track and meets business requirements.

QUESTION 1011

What is 'data mining'?

Answer

Data mining involves using various techniques to discover patterns and insights from large datasets, often employed to identify trends and relationships.

QUESTION 1012

What are the implications of Type 2 changes in customer dimensions?

Answer

Type 2 changes allow organizations to track historical changes by adding new rows for updated attributes, facilitating analysis of customer behavior over time while avoiding overcounting.

QUESTION 1013

What is meant by 'policy transaction amount'?

Answer

The policy transaction amount refers to the numeric value associated with a specific policy transaction, which can vary based on the transaction type.

QUESTION 1014

How is the relationship between a promotion and product sales assessed?

Answer

The effectiveness of a promotion is assessed by measuring sales lift, analyzing time shifts, cannibalization, and overall market growth during the promotional period.

QUESTION 1015

What is a 'Customer Relationship Management (CRM)' system?

Answer

A Customer Relationship Management (CRM) system is a technology used to manage a company's interactions with current and potential customers, streamlining processes and improving profitability.

QUESTION 1016

What are conformed dimensions?

Answer

Conformed dimensions are standardized dimensions that maintain consistency across different data marts, ensuring that they have the same keys, names, definitions, and values, allowing for coherent cross-mart analysis.

QUESTION 1017

What is the purpose of tracking inventory transactions?

Answer

Tracking inventory transactions provides detailed insights into the frequency and nature of inventory movements, helping to optimize inventory management and performance.

QUESTION 1018

What is the importance of effective and expiration dates in customer dimensions?

Answer

Effective and expiration dates are crucial for accurately tracking historical customer information, ensuring that analyses reflect the state of customer attributes at specific points in time.

QUESTION 1019

Define fact and dimension tables.

Answer

Fact tables store quantitative data for analysis, while dimension tables store descriptive attributes related to the facts for contextual understanding.

QUESTION 1020

What are slowly changing dimensions (SCDs)?

Answer

SCDs are dimensions that change over time, and they require specific techniques to manage updates without losing historical accuracy.

Flashcards 1021 - 1032

QUESTION 1021

What is a surrogate key?

Answer

A surrogate key is a type of primary key that is not derived from application data. It is usually an integer and does not have any intrinsic meaning, often used in data warehouses to manage slowly changing dimensions and missing data.

QUESTION 1022

How can organizations track changes in employee profiles?

Answer

Organizations can use a type 2 slowly changing dimension to track changes by creating new rows for each employee transaction while retaining historical data.

QUESTION 1023

What does behavior analysis refer to in the context of data warehousing?

Answer

Behavior analysis involves examining various customer and product behaviors, such as purchasing and payment patterns, to gain insights into business performance.

QUESTION 1024

Describe the type 1 approach to handling slowly changing dimensions.

Answer

Type 1 involves overwriting the previous dimension attribute value with the most current value, thus losing historical data.

QUESTION 1025

What does TCP/IP stand for?

Answer

TCP/IP stands for Transmission Control Protocol/Internet Protocol, which is the fundamental communication protocol of the Internet, consisting of a transport layer (IP) and an application layer (TCP).

QUESTION 1026

How does procurement impact an organization’s bottom line?

Answer

Effective management of procurement performance can significantly reduce costs and improve profitability, making it a crucial aspect of business operations.

QUESTION 1027

How is the promotion coverage fact table structured?

Answer

It contains one row for each product on promotion per store per day/week, allowing analysis of promotions independent of actual sales.

QUESTION 1028

What is the relationship between sales organization attributes and customer ship-to dimensions?

Answer

Sales organization attributes can either be combined with the customer ship-to dimension if highly correlated, or kept separate if their relationships are more complex and varied.

QUESTION 1029

What are dimensions in dimensional modeling?

Answer

Dimensions are descriptive attributes that provide context to the facts, helping users analyze data in various ways. Examples include date, product, customer, and promotion.

QUESTION 1030

What are the characteristics of a well-defined data mart?

Answer

A well-defined data mart is organized around business processes, integrates with the overall data warehouse architecture, and provides consistent data access across departments.

QUESTION 1031

What does 'scope creep' refer to in a data warehouse project?

Answer

Scope creep refers to the accumulation of minor changes in project requirements, which can lead to significant alterations in the project’s scope if not managed.

QUESTION 1032

What is the significance of the POS system in retail?

Answer

The POS system captures customer purchases, providing critical data for analyzing sales performance and customer behavior.

Flashcards 1033 - 1044

QUESTION 1033

What key factors affect human resources management data analysis?

Answer

Key factors include employee demographics, performance metrics, recruitment data, and training programs, which are crucial for workforce planning and development.

QUESTION 1034

What is the purpose of invoice transactions in a data warehouse?

Answer

Invoice transactions serve as a comprehensive record of customer shipments and financial details, providing critical insights into profitability and customer behavior.

QUESTION 1035

What is the purpose of a Date Dimension?

Answer

To provide a comprehensive representation of dates that includes various attributes like fiscal periods and holidays, facilitating time-based analyses.

QUESTION 1036

What is the significance of having a single, integrated view of each customer in CRM?

Answer

A single, integrated view of each customer allows organizations to coordinate their marketing, sales, operations, and service efforts effectively, enhancing customer interactions.

QUESTION 1037

What types of analyses can be performed with the budgeting chain?

Answer

Analyses can include comparisons of current commitments versus budgets, tracking changes in budget line items, and aggregating financial data over time.

QUESTION 1038

Describe the data warehouse bus architecture.

Answer

The data warehouse bus architecture provides a structured framework for building data marts in an incremental manner, allowing different processes to plug into a standardized data warehouse.

QUESTION 1039

What is a hybrid slowly changing dimension technique?

Answer

Hybrid techniques combine different slowly changing dimension approaches to provide flexibility while managing complexity, allowing for nuanced historical data tracking.

QUESTION 1040

Explain 'Type 1' Slowly Changing Dimension (SCD).

Answer

Type 1 SCD is a technique where the changed attribute is overwritten, meaning historical data is not preserved.

QUESTION 1041

What are the implications of ongoing technology advances for data warehousing?

Answer

Rapid advances in computing and communications technology require data warehouses to adapt continuously, ensuring they meet evolving user expectations and address security and privacy concerns.

QUESTION 1042

How do commitments relate to budgeting?

Answer

Commitments represent the financial obligations made against budget line items, helping organizations monitor spending against approved budgets.

QUESTION 1043

What challenges are associated with destructive updates in accumulating snapshot fact tables?

Answer

Destructive updates can lead to row width growth in databases, which may cause fragmentation and performance issues if not managed properly.

QUESTION 1044

What are the potential issues with generalizing cost data in a P&L statement?

Answer

Generalizing cost data can obscure specific profitability insights and lead to inaccurate assessments of financial performance across products or customers.

Flashcards 1045 - 1056

QUESTION 1045

What is the balance transaction fact table?

Answer

The balance transaction fact table records each transaction against an account, including the transaction amount and resulting balance.

QUESTION 1046

How can the concept of a value circle be applied in health care?

Answer

A value circle illustrates the interconnectedness of various stakeholders in health care, emphasizing the need for shared data for effective treatment and billing.

QUESTION 1047

What are the components of a data warehouse?

Answer

The main components include operational source systems, the data staging area, the presentation area, and data access tools.

QUESTION 1048

What is a hybrid approach to gathering requirements?

Answer

A hybrid approach combines individual interviews for detailed insights with facilitated sessions to achieve group consensus on requirements.

QUESTION 1049

What is meant by the term 'derived attributes' in the context of dimensions?

Answer

Attributes that are calculated based on other data points, often used to enrich dimension tables for more comprehensive analysis.

QUESTION 1050

How does a business define a trip in airline data?

Answer

A trip is defined as the complete journey taken by a passenger, which may consist of multiple segments and legs, typically starting and ending at the same airport.

QUESTION 1051

What role does senior management play in data warehouse integration?

Answer

Senior management's support is vital for fostering organizational consensus on integration, ensuring that data warehouse implementation aligns with business objectives.

QUESTION 1052

What is the purpose of the 'sales channel dimension' in airline data?

Answer

The sales channel dimension identifies how tickets were purchased, whether through travel agencies, direct airline sales, or online services, facilitating performance analysis.

QUESTION 1053

How do conformed dimensions benefit the health care consortium?

Answer

Conformed dimensions allow various organizations to share and analyze data seamlessly across different systems, ensuring consistency and accuracy.

QUESTION 1054

What are the primary core processes of an insurance company?

Answer

The core processes include issuing policies, collecting premium payments, and processing claims.

QUESTION 1055

What is the purpose of the lifecycle data track in data warehousing?

Answer

The lifecycle data track focuses on managing data throughout its lifecycle, ensuring quality and consistency from extraction to analysis.

QUESTION 1056

What is an accumulating snapshot in order management?

Answer

An accumulating snapshot is a fact table that tracks the progress of an order from initiation through shipment and invoicing, capturing multiple stages in a single schema.

Flashcards 1057 - 1059

QUESTION 1057

How can businesses utilize customer segmentation?

Answer

Businesses can utilize customer segmentation to identify distinct groups within their customer base for targeted marketing, improved service, and tailored product offerings.

QUESTION 1058

What is the purpose of the Country-Specific Date Outrigger?

Answer

To accommodate different national calendars and holidays, allowing multinational businesses to manage diverse date attributes effectively.

QUESTION 1059

How does management use data from the POS system?

Answer

Management analyzes POS data to understand sales patterns, customer preferences, and the effectiveness of promotions.